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AAFL — An Agent-Augmented Framework for Learning

Judgment, Gates, and Workplace Performance in the Agent Era

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Foreword: status and provenance

AAFL is an independent framework. It is offered to the field as a working artifact: opinionated, citable, and revisable. The framework's intellectual debts are deep and broad; they are named in §9 as a substantive lineage rather than as plank-by-plank derivations, because AAFL is intended to read as a single synthesis, not a composite of borrowed parts. Critique should engage the framework on its own terms.

This is v1.3 (2026-05-20).

Executive Summary

Instructional design has spent fifty years answering the question "*how do we build courses that learners pass?*" The performance-oriented corners of the field have spent decades quietly reframing the question to "*how do we accelerate workplace performance?*", under the broader tradition of Human Performance Technology and through specific operational articulations like Action Mapping, 5 Moments of Need, and the Success Case Method. AAFL extends that workplace-performance tradition into the agent era; the framework is positioned as an integration of established HPT, instructional-design, and contemporary AI-in-instructional-design work, not as a clean-break theoretical contribution.

The agent era forces a third reframing. As AI agents become competent first-draft authors, capable of producing learner personas, learning objectives, storyboards, item banks, and accessibility-checked content at the level of a mid-career instructional designer, the human ID practitioner's center of gravity shifts from *production* to *judgment*. The skill that compounds is no longer *generating instructional artifacts*; it is *deciding which of three agent-generated drafts is pedagogically defensible, why, and what the agent missed about the learner.*

This whitepaper introduces AAFL — the Agent-Augmented Framework for Learning: a framework for the era of agent-authored first drafts and human-held accountability. AAFL preserves ADDIE's five-phase spine (because federal contracting officers, ACE evaluators, and Quality Matters reviewers still read in that vocabulary), adds three cross-cutting layers (orchestration, evaluation-as-spec, governance) that did not exist in classic ADDIE, introduces an inner *Translator's Loop* every artifact runs through, and is anchored in **workplace performance**, not course completion or learning attainment, as the organizing outcome.

AAFL formalizes five intellectual positions that no published competitor currently integrates:

1. **An HITL decision-gate taxonomy**: eight named gates with explicit pedagogical-vs-production classification
2. **A pedagogical vs production task taxonomy**: agents do what is bounded and reviewable; humans hold what is value-loaded and trust-bearing

3. **The Proportional Restraint Scale (PRS)**: a four-level scale (PRS-1 Sandbox · PRS-2 Co-designer · PRS-3 Production Partner · PRS-4 Performance System) with prescribed ceilings for each consequence space; most federal and accredited programs sit at PRS-2 or PRS-3 (see §6)
4. **A performance-anchored eval framework**: six dimensions with workplace performance attainment as the primary spine; eval rubric authored at the Performance Outcome Gate before any agent runs
5. **A governance/escalation layer**: the institutional infrastructure (audit trail, classification handling, accreditation alignment, escalation paths) that operationalizes the PRS ceilings; the federal-applicability claim no civilian framework currently makes

The framework is opinionated about its own limits. It publishes what it *won't* do (concrete caps on agent autonomy tied to data classification, training-program type, and decision class) before publishing what it does. That posture is the source of its credibility, not its limitation.

This whitepaper is structured to be read in two ways. **As a sequential argument**, it moves from the conceptual realignment (§1), through the architecture (§2–§3), the gate taxonomy (§4), the agent-task split (§5), the Proportional Restraint Scale (§6), the eval framework (§7), the cross-cutting layers (§8), engagement with prior frameworks (§9), federal applicability (§10), risks (§11), and limitations (§12). **As a reference manual**, each section is self-contained: practitioners can use the gate taxonomy or the eval framework à la carte without adopting the whole framework.

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§1. The Conceptual Shift

Florida State University's 1975 *Interservice Procedures for Instructional Systems Development*, the genealogy of what became ADDIE, assumed a clear sequence of human authorship. The instructional designer analyzed learner needs, designed learning objectives, developed content, implemented delivery, and evaluated results. Each phase had a human author. Tools were accelerants; they did not change the locus of authorship.

The agent era reverses who writes the first draft without reversing who is accountable for it. Agents now produce drafts that human practitioners used to author by hand; institutional, legal, and pedagogical accountability for the work still rests with the practitioner. This is *authorship inversion without accountability inversion*. The *who-writes* role flips; the *who-owns-it* role does not.

A working definition. Throughout this paper, an *agent* is a generative AI system (at minimum a large language model, more typically an LLM coupled to retrieval, tool use, and a controller that decomposes a brief into multi-step work) operated against an instructional-design task with sufficient autonomy to produce a complete first-draft artifact (a persona, an objective set, a storyboard, an item bank, an analytics interpretation) without per-step human authoring. The definition deliberately spans a range: a single prompted model answering a structured brief sits at the low end; a multi-step orchestration that pulls SME transcripts, drafts content, runs a WCAG scan, and packages a SCORM bundle sits at the high end. AAFL's gates and PRS levels are written to hold across the range. Where the framework's claims depend on a specific end of the range (for example, the PRS-3 and PRS-4 prescriptions assume an orchestrated multi-step agent, not a single chat turn) the text says so. "Agent" in this paper is not a marketing term for any one product; it is the operational unit of work the framework governs.

A competent AI agent, given clear instructions, well-formed inputs, and access to source material, can now produce:

- A learner persona drafted from interview transcripts in minutes
- A set of objectives at multiple cognitive levels
- A backward-mapped task list grounded in workplace performance metrics
- A storyboard with branching scenarios
- An item bank with distractors and rubrics
- Accessibility-checked content meeting WCAG 2.1 AA
- A facilitator guide and instructor enablement package
- Learning analytics dashboards with hypothesized causes for learner outcomes

This is not aspirational; it is operational. Across commercial authoring platforms in 2024–2026, design-loop AI tooling has moved from optional add-on to default surface. Course-outline generation, narrated-video production at scale, animated scenario-based training, and accessibility-checked content drafting are now common features of mainstream instructional design platforms. Industry-level adoption signals from the Association for Talent Development's 2025 State of the Industry Report (ATD Research, 2025) show 55% of surveyed organizations providing AI practical skills training in 2024, with 75% expecting to increase AI spending the following fiscal year — directionally consistent with the platform-level capability shift practitioners are operating inside. On the federal side, in the author's federal ID practice, AI-Enhanced ADDIE workflows applied across ACE-accredited graduate certificate and post-baccalaureate programs have produced material first-draft cycle compression, with practitioner-observed reductions typically falling in the 40–60% range across the deliveries the author has direct visibility into. The measurement frame is ID-team hours from brief acceptance to first-draft sign-off, relative to comparable pre-2024 baselines on adjacent programs in the same portfolio; the underlying figures are practitioner-level observations across multiple production deliveries (exact denominators withheld for client-confidentiality reasons), not controlled-study results; see §12 (Limitations and Future Work). The capability claim is not in dispute.

What *is* in dispute, and what AAFL claims to resolve, is the question of where the human stays, why, and what authority the human's judgment carries that the agent's output does not. Two positions in the discourse currently dominate:

- **The substitution position.** Agents will replace ID practitioners. Most extreme cases: per-learner curriculum generated end-to-end without human review.
- **The augmentation position.** Agents accelerate ID work without displacing it. Humans review agent output and make the value-laden decisions.

AAFL takes the augmentation position. The substitution argument deserves a fair hearing before it is set aside, because some of its underlying motives are legitimate: consistency at scale (an autonomous agent does not have a bad day, a busy week, or an idiosyncratic eye for one learner persona over another), removal of certain known biases that affect human reviewers, and per-learner unit economics that human-led design cannot match in high-volume contexts. AAFL does not claim that substitution is impossible or undesirable in every context; it claims that the contexts the framework addresses (federal training, accredited and credentialing programs, workplace-performance contexts where institutional accountability is non-transferable) have audit, certification, classification, and accountability structures that substitution cannot satisfy. A credential authored end-to-end by an autonomous agent cannot defend itself in accreditation review; a CUI training program authored without human classification review cannot ship; a high-consequence performance intervention whose causal claim no human will defend in front of a stakeholder is not an intervention an institution can operate. The framework's narrow claim is about *these* contexts, not about every L&D context everywhere.

Inside the augmentation position, "augmentation" is itself broad. *How* the human relates to the agent's output is what separates augmentation that works from augmentation that drifts into rubber-stamp review. AAFL calls this practitioner posture the **translator stance**: agents produce confident first drafts; humans translate between agent capability and instructional soundness, judging at specific decision points, gating release, and holding the accountability the agent cannot. The translator stance is not a third position on the spectrum; it is the discipline that operationalizes the augmentation position. Substitution requires no stance, because there is no human to take one. Augmentation without the translator stance collapses into review-theater.

The translator stance holds that the *authorship inversion* is real (agents write first, humans validate) but that the *accountability inversion* must not be. The learner does not care who wrote the artifact. The learner cares whether the artifact improves their performance. The institution does not care who drafted the assessment. The institution cares whether the assessment validly measures the construct claimed and whether it stands up in audit. Agents do not (yet) bear professional, legal, or institutional accountability for these claims; humans do. AAFL formalizes the gates where human accountability is irreducible regardless of agent capability.

But the conceptual realignment goes deeper than authorship. The *purpose* of doing ID is changing, too. More accurately, the agent era exposes a change that was already underway in the most performance-oriented corners of L&D, and provides the eval tooling to finally complete it.

ADDIE was built around *learning attainment*: phases optimize for producing a course that learners pass. That framing was not a failure of imagination on the part of its architects. The performance-oriented tradition recognized the gap from the beginning. Thomas Gilbert, Robert Mager, Geary Rummler and Alan Brache, and the broader Human Performance Technology field that grew up around their work argued that workplace performance, not course completion, was the legitimate outcome of L&D investment. The constraint was *tooling*. Pre/post tests, knowledge checks, retention assessments, multiple-choice quizzes: these were what could practically be measured at scale through the 1980s, 1990s, and 2000s. The eval tools of those decades let practitioners ask *did the learner absorb the content*, but not *did the absorption change what they do at work tomorrow*.

The agent era is beginning to relax that constraint. AI agents, paired with augmented-reality overlays in some industrial-maintenance and field-service contexts, instrumentation in increasingly digitized work environments, and per-learner telemetry from learning-platform integrations, are starting to make *while-you-work* evaluation possible at meaningful scale in bounded domains: did this technician fix the machine faster? Did this analyst flag the right indicator under load? Did the credentialed officer execute the correct procedure during the incident? The measurement is, in those domains, beginning to keep up with the outcome question. This is not a uniform capability. AR-overlay tooling at the point of work is operational today primarily in industrial and field-service settings; per-learner workflow telemetry of the fidelity Gate 8 would prefer remains uneven across federal training contexts, knowledge-work environments, and credentialing programs. What learning scientists could only argue for in the abstract is becoming operationally available *where the instrumentation already exists*; AAFL treats that asymmetry honestly in §12 (Limitations).

That is why the choice of organizing outcome is no longer abstract; it is the spec the agent fulfills. An agent will optimize whatever you tell it to optimize. If the eval rubric measures course completion and quiz scores, the agent will produce courses that maximize completion and quiz scores, and the result may be entirely uncorrelated with workplace performance. If the eval rubric measures time-to-competency and transfer-to-job, the agent's output orients toward those outcomes instead.

AAFL therefore takes the opinionated stance: **workplace performance is the framework's organizing outcome, not learning attainment.** Every gate, every eval dimension, every PRS level ties back to performance. ADDIE's five-phase spine survives (practitioners still Analyze, Design, Develop, Implement, and Evaluate) but the organizing question those phases answer changes from *did the* *learner pass* to *did workplace performance change*. AAFL is continuity-with-evolution: it inherits ADDIE's process, refits it for agent-era eval tools, and anchors it where the HPT tradition always argued the outcome belonged. The substantive engagement with named voices is consolidated in §9.

The realignment, in summary, is from:

- *Designer as maker → designer as judge, gate-keeper, and accountability holder for the system that makes*
- *Course completion as the win → workplace performance as the win*
- *AI as accelerant for human authorship → AI as first-draft author within human-defined judgment gates*
- *Eval tooling as the limit on what could be measured → eval tooling as the spec the agent fulfills*

What follows is the architecture that operates inside this realignment.

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§1.5. Method — How This Framework Was Developed

A framework paper that proposes a new operating model for a field's practice owes the reader an account of how the proposal was arrived at. AAFL was developed as a **practitioner-researcher synthesis**, informed by the design-based research and design-based theorizing traditions (Edelson, 2002; McKenney & Reeves, 2013; McKenney & Reeves, 2018) without claiming the full apparatus of those methodologies (iterative cycles of intervention testing with documented design conjectures and refinement traces). The synthesis is grounded in two substantive inputs: a literature scan of the human performance technology and agent-era AI-in-instructional-design (AI-in-ID) traditions, and the author's federal instructional design practice across multiple production deliveries. Calling it design-based *theorizing* rather than design-based *research* is deliberate, following Edelson (2002): the framework is a theoretical proposal advanced for the field to test, not the output of a completed multi-cycle research program (see §12 Limitations).

Inputs to the synthesis

Literature scan (2025–2026). A bounded review of three intersecting bodies of work: (i) the human performance technology (HPT) tradition (Gilbert, 1978; Rummler & Brache, 1990; Moore, 2017; Gottfredson & Mosher, 2011; Brinkerhoff, 2003; Kirkpatrick & Kirkpatrick, 2016) for the *workplace performance as organizing outcome* commitment; (ii) the contemporary AI-in-ID literature (Chai et al., 2025; Hardman, 2025; EduPlanner — Zhang et al., 2025; AUSS — Arya Mary et al., 2026) and the practitioner-competency literature surrounding it (Dakan & Feller, 2023–2025, anchored in the *AI Fluency for Educators* and *AI Fluency: Framework & Foundations* courses published by Anthropic Academy) for the agent-era state of the art; and (iii) methodological adjacencies, namely eval-driven agent engineering practice (Anthropic Engineering, 2026), human-algorithm centaur models (Saghafian & Idan, 2024), and empirical work on uneven AI capability across tasks (Brynjolfsson et al., 2023; Dell'Acqua et al., 2023). The positioning records, white-space analysis, named-voice engagement map, and full annotated reading list are versioned alongside this whitepaper in a public project repository (see §1.5 Reproducibility below).

Practitioner experience. The author has applied AI-Enhanced ADDIE workflows across ACE-accredited graduate certificate and post-baccalaureate programs in federal security education. Adjacent practice that informed the framework's design choices, particularly around the human-in-the-loop discipline and the eval-as-spec stance, includes extensive AI-assisted software-engineering work; advocacy-research and policy-paper authorship for an unnamed civil-society organization, where AI-drafted materials had to clear external-audience credibility bars; and a sustained curriculum-design body of work for an unnamed performing-arts credentialing program operating internationally, where the agent-era split between source-truth fidelity and pedagogical strategy first surfaced as a recurring design problem. The federal practice produced the operational observations on first-draft cycle compression reported in §1 and the practitioner-level PRS-1 / PRS-2 productivity bands reported in §6. These are practitioner-level observations across multiple production deliveries; they are not the outputs of a controlled study, and the framework treats them as such (see §12, Limitations and Future Work).

Synthesis method

The framework's distinctive choices (the eight HITL gates, the pedagogical-vs-production split, the Proportional Restraint Scale with classification caps, the eval-as-spec discipline, and the three cross-cutting layers) emerged from this synthesis under four design criteria: (a) preserve the workplace-performance commitment of the HPT tradition, (b) absorb the eval-driven and centaur-model methodological adjacencies, (c) extend the named published AI-in-ID frameworks rather than competing with them, and (d) address the federal-applicability white space that no civilian framework currently occupies. The integration is the contribution; no individual element is novel in isolation.

What this method is *not*

AAFL has not been developed via external expert-panel consensus methods such as the Delphi technique (the iterative-expert-feedback methodology common in education-research framework development); it has not been validated by independent field tests at PRS-3+; it does not draw on federally-funded user research; and it does not yet rest on published case studies that any reader can audit. These are deliberate scope boundaries for this whitepaper: the framework is offered to the field as a *spec inviting field validation*, not as a validated framework. The §12 Limitations and Future Work section names this explicitly. Field tests, external review panels, and comparative practitioner studies are the natural next steps; the framework's structure tolerates row-level updates without changing its architecture (see §11, Risk 4, and §5 "When the line moves").

Reproducibility

The literature scan, white-space analysis, named-voice engagement map, and source-citation bibliography are versioned alongside this whitepaper in the AAFL project repository at `github.com/ruchir-code/aafl-whitepaper`. A reader who wishes to audit the synthesis can trace any claim in AAFL back to its source in those documents. The repository carries the same proprietary license as the whitepaper; the materials are made available for reader audit, citation, and academic engagement under the terms in `LICENSE`.

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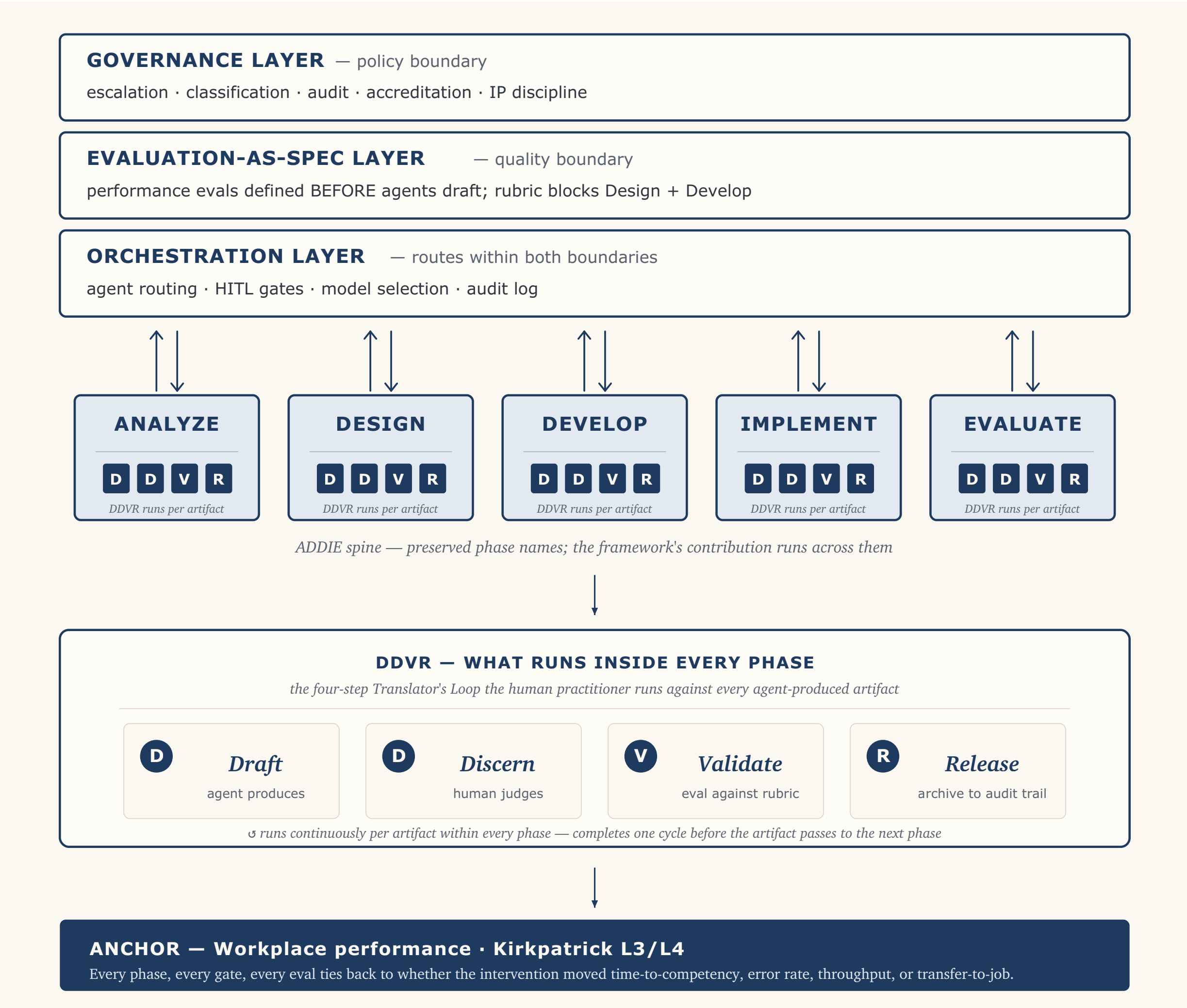
§2. Architecture — The Hybrid Stack

AAFL preserves ADDIE's five phase names as the **vertical spine** and introduces three **cross-cutting horizontal layers** that did not exist in classic ADDIE.

The decision to keep ADDIE's vocabulary is deliberate. Federal contracting officers, ACE evaluators, Quality Matters reviewers, military training command staff, and corporate L&D leadership all read in ADDIE's vocabulary. A framework that begins by replacing the vocabulary forfeits the audience most likely to adopt it. AAFL assumes the audience knows ADDIE; the framework's contribution is what runs *across* the phases, not what replaces them.

The hybrid stack



> **Figure 1 — alt-text caption.** *AAFL framework architecture. Three horizontal cross-cutting layers stacked top to bottom: Governance, Evaluation-as-Spec, and Orchestration (each layer's specific responsibilities are expanded in the What's in each cross-cutting layer subsection below). Beneath the three layers, ADDIE's five phases (Analyze, Design, Develop, Implement, Evaluate) run left to right as the framework's vertical spine. Each phase connects to the three layers above via bidirectional arrows: the layers shape the phase's work, and the phase reports back into the layers' audit, eval, and governance state. Each phase column carries an embedded DDVR indicator (four small chiclets reading D-D-V-R) signaling that the four-step Translator's Loop (Draft → Discern → Validate → Release) runs within each phase against every agent-produced artifact. A legend below the phase row expands DDVR into its four numbered steps with their human-or-agent role assignments. At the bottom sits the framework's anchor: workplace performance, Kirkpatrick L3/L4. Every phase, every gate, every eval ties back to whether the intervention moved time-to-competency, error rate, throughput, or transfer-to-job, not whether the learner passed.*

Why hybrid (not pure overlay, not pure replacement)

Three architectural choices were considered:

- **(a) Pure overlay.** Keep ADDIE phases unchanged, bolt on HITL gates as checklists. This is too conservative. Orchestration, evaluation, and governance need to run *continuously* across every phase; they cannot be a Phase 6.
- **(b) Pure replacement.** Discard ADDIE phases, introduce new phase names like *Brief → Generate → Discern → Validate → Release → Learn*. Too radical. Forfeits ADDIE Guide lineage, federal vocabulary, and accreditation language. Reduces credibility with the audience most likely to adopt.
- **(c) Hybrid.** Preserve the phase names, add cross-cutting layers and an inner motion. Practitioners keep the map they already use; AAFL adds the new instruments running across it.

AAFL adopts (c). ADDIE Guide remains the *reference* (the dictionary); AAFL is the *operating manual* (the process for using the dictionary in the agent era).

What's in each cross-cutting layer

The layers are presented here in the same top-to-bottom order they appear in Figure 1: Governance (the outermost policy boundary), then Evaluation-as-Spec (the quality boundary inside it), then Orchestration (the runtime within both). The nesting is deliberate. An orchestration decision cannot violate the evaluation boundary above it, and an evaluation decision cannot violate the governance boundary above that. Reading from the top down also matches how an institution actually adopts the framework: governance is the first thing legal and procurement teams ask about; eval discipline is what L&D leads commit to next; orchestration is the runtime concern that comes last because it depends on the boundaries above it being settled.

The Governance Layer is the institutional infrastructure. It captures audit trail (chain of custody for every agent-generated artifact, model version recorded, prompt-and-response archived for the contract retention window), classification handling (FedRAMP, NIST 800-171, CUI marking, ITAR/EAR where applicable), accreditation alignment (ACE evaluator expectations, Quality Matters Standards 2/3/8 (Quality Matters, 2023), WCAG 2.1 AA + Section 508 (W3C, 2018)), escalation paths (what happens when the human reviewer disagrees with the agent's output, when does human override, when does the disagreement escalate), and **PRS caps by data classification**: the federal-applicability claim no civilian framework currently makes.

The Evaluation-as-Spec Layer borrows the eval-driven discipline from agent engineering practice ("define evals before agents fulfill them"; Anthropic Engineering, 2026) and operationalizes it for ID. The eval rubric for a given course is authored at the Performance Outcome Gate (Gate 2) *before* any Design-phase or Develop-phase agent runs. The rubric *blocks both Design and Develop*. No rubric, no Design, and certainly no Develop. The reasoning is straightforward: a malformed design is worse than no design, because malformed designs commit reviewers, schedule, and downstream production to the wrong target. This is the framework's single biggest defense against the agent-era failure mode of artifacts that are well-formed but not instructionally sound.

The Orchestration Layer routes work between agents and humans (Anthropic Research, 2024, on orchestrator-workers patterns offers a useful technical articulation of the runtime substrate AAFL's orchestration layer assumes). It decides which agent runs which task, which model is appropriate for which class of work (frontier vs cost-optimized vs federal-tenant), when artifacts route to which gate, and what gets logged. At PRS-1, orchestration is manual: the human is the orchestrator. At higher PRS levels, orchestration becomes increasingly automated within the boundaries Governance and Evaluation-as-Spec define.

The three layers work together in nested order: Governance sets the policy boundary; Evaluation-as-Spec sets the quality boundary inside it; Orchestration routes work within both.

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§3. The Translator's Loop

The translator stance (§1) is the practitioner's posture toward agent output; the Translator's Loop is the cycle the stance runs against every artifact. A stance is what the human ***is****; a loop is what the human* ***does****.*

Inside every phase, every artifact runs through a four-step inner cycle. The diagram-side mnemonic is **DDVR**, the four chiclets embedded in each ADDIE column of Figure 1:

> ***Draft → Discern → Validate → Release***

This is *the unit of work* in AAFL. Practitioners measure progress not in deliverables but in loop completions. A phase ends when all its artifacts have passed the loop.

Draft. The agent generates the artifact against a brief that includes the eval rubric authored at the Performance Outcome Gate. The brief is precise: it specifies the artifact type, the source material, the constraint set (cognitive level, modality, accessibility tier, classification), and the eval criteria.

Discern. The human applies the appropriate HITL gate criterion (see §4). The Anthropic AI Fluency framework (Dakan & Feller, 2023–2025) gives the practitioner-side competency that operates here: *Discernment* is the act of judging which agent output meets the criterion.

Validate. For production-class artifacts, a second pass against eval criteria: automated where possible (accessibility scans, citation checks, classification marking); human where not (pedagogical fit, learner-context judgment).

Release. The artifact is signed by the accountable human, archived to the audit trail (Governance Layer), and made available to downstream phases.

The Translator's Loop is also the practitioner-facing slogan for AAFL. Where the framework name (AAFL) is institutional and acronym-coded, the Translator's Loop is human and process-coded. Both terms operate in the literature; both refer to the same framework.

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§4. The HITL Decision-Gate Taxonomy

Eight gates. A memorable count, mapped to ADDIE phases, with each gate classified as either **pedagogical** (humans-only, value-statement decisions) or **production** (agent-prepared, human-signed, sampleable at higher PRS levels). The split is the framework's load-bearing distinction.

Gate 1 — Learner Reality (Analyze, pedagogical)

Stake. Whether the design serves a real population or a synthetic one.

Agent input. Drafted learner persona; prior-knowledge map; performance-gap hypothesis; demographic and contextual analysis.

Human criterion. Does this match what I have actually heard from this audience? What is the agent confidently wrong about?

Failure mode. Designing for the LLM's median-case learner instead of the actual workplace population. Ships a course that solves no one's problem.

Gate 2 — Performance Outcome (Analyze → Design boundary, pedagogical)

Stake. The terminal definition of mastery: what workplace performance looks like when the intervention has worked. This is where AAFL departs from legacy *learning objectives*. Traditional learning objectives describe what a learner should know or be able to do at the end of the course; AAFL's performance outcomes describe what the learner should *do differently at work*. The two are not interchangeable. A learner can master every learning objective and still not change a single workplace behavior, which is precisely the gap the pre-agent eval tooling left open (see §1). This gate is where the new eval tooling has the most to offer: *while-you-work* signals make these performance outcomes practically measurable for the first time.

Agent input. Candidate objectives at multiple cognitive levels; backward-mapped action lists; performance-context scenarios; assessment criteria.

Human criterion. Are these the performance outcomes that matter, or the outcomes that are easiest to measure? Time-to-competency, error rate, throughput, transfer-to-job: defined and measurable?

Failure mode. Defaulting to learning-attainment proxies (course completion, knowledge-check scores) instead of performance metrics. These proxies persisted because the eval tools of the pre-agent era could not reliably measure anything else. The agent era removes that excuse, and an agent will optimize whatever you tell it to optimize, including the wrong thing.

Critical role. This gate **blocks** the Evaluation-as-Spec Layer for both the Design and Develop phases. The eval rubric is authored here, *before* any Design-phase or Develop-phase agent runs. No rubric, no Design, no Develop. The framework's working principle is that a malformed design is worse than no design at all: a design committed to the wrong performance outcome will marshal downstream production effort, reviewer attention, and schedule around an incorrect target, and the cost of unwinding that is far higher than the cost of pausing Design until Gate 2 is signed.

The iterative-design counterargument, addressed. A reasonable ID reviewer will object that performance rubrics are themselves often *clarified* through design iteration: the storyboard reveals an ambiguity in the outcome, the prototype surfaces a measurement problem the rubric didn't anticipate, and forcing a complete eval rubric before any design work begins risks producing a rubric that is precise but wrong. The objection has merit in a pre-agent workflow, where design iteration is slow and expensive enough that an imperfect rubric refined-as-you-go is the rational tradeoff. The agent era changes the math. When a Design-phase agent can produce a complete storyboard in hours, the cost of generating that storyboard against the wrong outcome (and then regenerating against a corrected one) is low; the cost of *committing reviewer attention and stakeholder expectations* to the wrong-outcome storyboard is not. Gate 2's blocking discipline does not prohibit rubric revision; it requires that an *initial* rubric exist as a signed artifact before Design runs, so that subsequent Design-driven rubric refinement is a deliberate amendment to a committed spec rather than backfilled justification for whatever the agent produced. The discipline is *eval-first*, not *eval-frozen*.

Gate 3 — Pedagogical Strategy (Design, pedagogical)

Stake. Whether the instructional approach (scaffolding sequence, modality mix, practice-to-feedback ratio) fits the learner and the content.

Agent input. 2–3 candidate strategies with rationales; cognitive-load analysis; modality recommendations; UDL-fit notes.

Human criterion. Which of these survives contact with this specific learner population in this specific workplace context? Which one accelerates performance the soonest with the least cognitive cost?

Failure mode. Adopting a strategy because it is well-formed in the agent's draft rather than because it fits the learner.

Gate 4 — Source Truth (Develop, production)

Stake. Factual and SME accuracy of generated content.

Agent input. Drafted content with citations; SME-validated transcripts mapped to drafted points; hallucination-flag pass.

Human criterion. SME-validated, traceable to source, free of hallucinated authority. The conceptual point here is sharpened by Hicks, Humphries, and Slater (2024): the more honest framing is that LLM output is *bullshit* (Frankfurt's technical sense: indifference to truth) rather than *hallucination* (which implies a perceptual failure the model has and would correct if it noticed). Gate 4 is the structural response: the human practitioner's truth-tracking commitments are what the agent does not have.

Federal note. Non-negotiable in federal contexts. CUI/classification-marked sources only; classified material does not pass through general-purpose agents (caps at PRS-2).

Gate 5 — Equity & Accessibility (Develop, production)

Stake. WCAG/UDL conformance, representational fairness, cognitive load fit.

Agent input. Accessibility-checked draft; alt-text; transcripts; captions; reading-level analysis; automated WCAG scan results; representational-bias audit.

Human criterion. Does this work for the learner whose context I have least imagined?

Gate 6 — Assessment Validity (Develop, production)

Stake. Whether the assessment measures the stated outcome.

Agent input. Item bank; rubric; distractor analysis; item-response statistics where available.

Human criterion. Construct validity, not surface plausibility. Does a learner who passes this actually have the competency that accelerates workplace performance?

Failure mode. Agent-generated items that look professional but measure surface fluency (recall) rather than the targeted competency (transfer). An assessment that does not predict workplace performance is theatre, regardless of psychometric polish.

Gate 7 — Release (Implement, production)

Stake. Whether the artifact is fit to put in front of learners.

Agent input. QA bundle; packaging (SCORM/xAPI/cmi5); deployment checklist; 508 conformance report; classification marking confirmation.

Human criterion. Named accountable owner. Reviewer signature. Artifact-of-record archived to the Governance Layer audit trail.

Gate 8 — Performance-Effect (Evaluate, pedagogical)

Stake. Whether the intervention actually accelerated workplace performance, not just whether the course was completed. "Accelerated workplace performance" is increasingly operationalized through *while-you-work* signal capture where the instrumentation exists: AR overlays at the moment of task execution that log whether the technician completed the procedure correctly the first time; instrumented work environments that record cycle-time and error rates; per-learner telemetry that surfaces the gap between training-environment competence and on-the-job execution. These signals, unavailable to ADDIE's original architects, are making Gate 8 operationally answerable in instrumented domains; in less-instrumented federal and credentialing contexts, Gate 8 remains answerable through the more traditional Kirkpatrick L3 instruments (supervisor observation, on-the-job performance review, post-training transfer studies) that the HPT tradition has long deployed. The framework does not require AR or telemetry; it requires *some* honest performance-outcome signal at Gate 8, with the fidelity calibrated to what the context can support.

Agent input. Outcome data; analytics; drop-off patterns; pre/post performance metrics; transfer-to-job indicators; error-rate change; time-to-competency analysis; hypothesis on causes.

Human criterion. Causal interpretation, decision to iterate, learner-harm assessment. Did performance get faster, more accurate, more capable, and if so, can we credibly claim the intervention caused it?

Vocabulary note. This is Kirkpatrick L3 (Behavior) and L4 (Results) territory in the L&D dialect every reviewer of this work will use. The framework's organizing outcome lives or dies here.

The gates at a glance

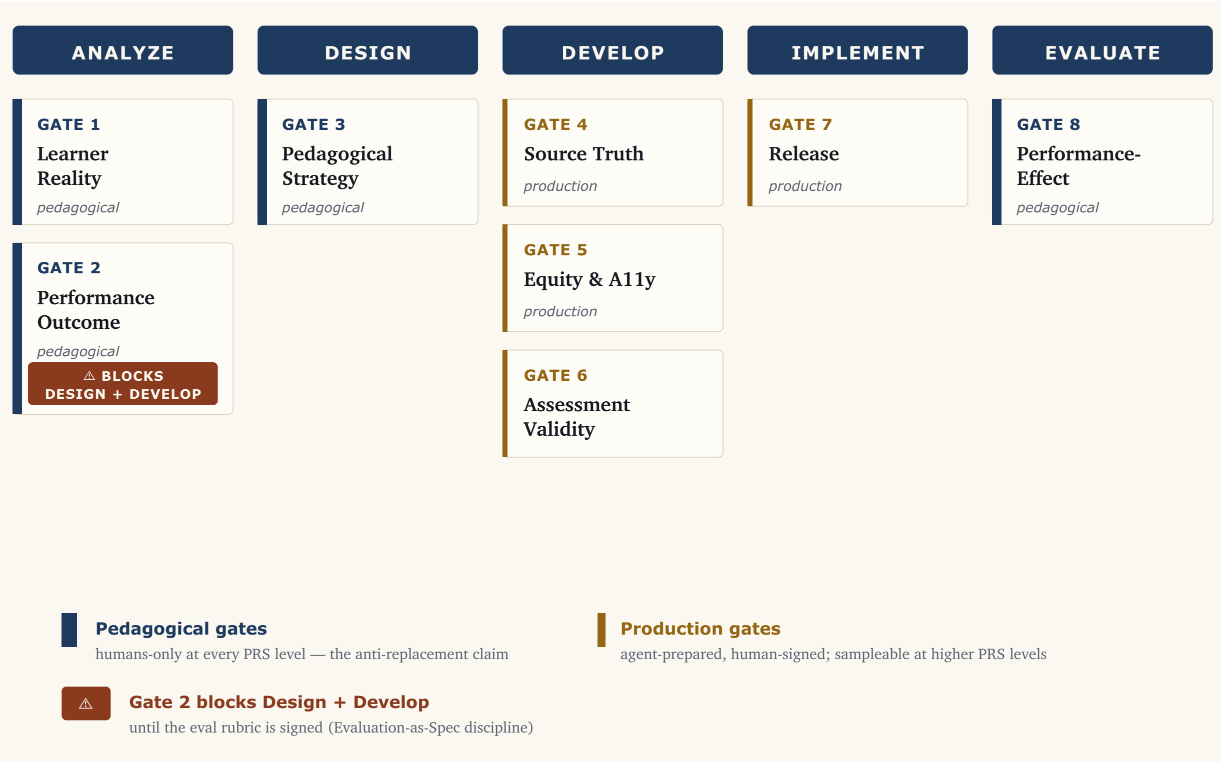

Figure 2 — alt-text caption. *The eight AAFL HITL decision gates mapped to ADDIE phases. Gate 1 (Learner Reality, pedagogical) and Gate 2 (Performance Outcome, pedagogical, blocks Design + Develop) sit in the Analyze phase. Gate 3 (Pedagogical Strategy, pedagogical) sits in the Design phase. Gates 4 (Source Truth), 5 (Equity & Accessibility), and 6 (Assessment Validity), all production gates, cluster in the Develop phase. Gate 7 (Release, production) sits in the Implement phase. Gate 8 (Performance-Effect, pedagogical) sits in the Evaluate phase. The four pedagogical gates (1, 2, 3, 8) bookend the framework and remain humans-only at every PRS level. The four production gates (4, 5, 6, 7) cluster in Develop and Implement and are agent-prepared and human-signed, becoming sampleable at higher PRS levels. Gate 2 is marked as the framework's structural blocker: until the eval rubric is signed at Gate 2, neither the Design nor the Develop phase can run.*

Reading the gate map

The first three gates and the last (Reality, Performance Outcome, Pedagogical Strategy, Performance-Effect) are *pedagogical*. They are humans-only at all PRS levels. They are where the framework's anti-replacement claim lives: these are the irreducibly human decisions, regardless of agent capability.

The middle four (Source Truth, Equity, Assessment Validity, Release) are *production*. They are agent-prepared and human-signed. At PRS-1 and PRS-2, every production-gate output is reviewed. At PRS-3, only flagged outputs trigger review (the agent's own confidence scoring + automated checks gate the human queue). At PRS-4, production gates become statistical sampling: humans audit a representative slice rather than every artifact.

The pedagogical gates do *not* relax at higher PRS levels. That asymmetry is what prevents the framework from sliding into pure agentic autonomy.

A note on why Gates 5 and 6 sit on the production side. A reasonable ID reviewer will object that both Gate 5 (Equity & Accessibility) and Gate 6 (Assessment Validity) carry decisions that look value-loaded: *which accommodations are reasonable for the learner whose context I have least imagined* is not a checklist question, and *whether a multiple-choice item measures the construct claimed or only surface fluency* is one of the most judgment-heavy decisions in instructional design. Classifying them as *production* (and therefore sampleable at higher PRS levels) appears to contradict the framework's own load-bearing distinction.

The answer is a sub-distinction inside the gates themselves. The *production-prep work* at each gate is genuinely automatable and reviewable: WCAG scans, alt-text generation, contrast checks, item-statistics computation, distractor analysis, reading-level adjustment. Sampling at PRS-3+ applies to *that work*: the agent's prep becomes statistically auditable rather than 100% reviewed. The *human criterion* at each gate, the unsampled commitment that the gate makes, remains a per-artifact human sign-off even at PRS-4. Gate 5's accommodation-reasonableness judgment is not sampled; Gate 6's cut-score setting and construct-validity decision are not sampled. The framework's claim is narrower than the production label suggests on first read: the *throughput at the gate* scales with the PRS level; the *load-bearing human decision the gate names* does not. A future revision may make this sub-distinction explicit at the gate definitions; for v1.3 the commitment is stated here for the record.

§5. The Agent-Task Taxonomy

The framework's deterministic answer to "what does the agent do, what does the human do," and the answer it gives to "will AI replace ID practitioners."

The split principle

> *Anything where the cost of being wrong is bounded and reviewable goes to the agent. Anything where being wrong harms a learner, breaks trust, or commits the institution stays with the human.*

The taxonomy

The taxonomy uses a third-column tag: **DL** (design loop: compresses ID cycle time), **PL** (performance loop: compresses learner time-to-competency or improves transfer), or **DL+PL** (both).

Reading the PRS prefixes. Untagged rows apply at all PRS levels. Rows prefixed **(PRS-3+)** mark capability available only at PRS-3 and above; rows prefixed **(PRS-4)** mark capability available only at PRS-4. The prefix is a PRS gate, not a row category: these are not separate "advanced" rows but specific capabilities that PRS-1/PRS-2 implementations should not attempt because the governance discipline required to operate them has not been established. See §6 (Proportional Restraint Scale).

Agent-led tasks	Human-only tasks	Loop
Drafting learner-persona templates from raw inputs	Deciding *who the real learner is* and what they don't yet know they need	DL
Pulling SME interview transcripts into structured analysis notes	Choosing which SME contradictions are signal vs noise	DL
Generating candidate learning objectives at multiple cognitive levels	Selecting which objectives define mastery for *this* program	DL
Mapping objectives to existing competency frameworks (ACE, NICE, OPM, JKO; see Appendix A for acronym expansions)	Defending the mapping in an accreditation review	DL
Drafting backward-mapped action lists from performance gap analysis	Validating that the actions actually move workplace performance	DL+PL
Generating workflow-moment scenarios from job task analysis	Confirming the moments map to real workflow behaviors	DL+PL
Drafting storyboards, scripts, scenario branches	Pedagogical sequencing; narrative coherence; learner-emotional arc	DL
First-pass content generation against a validated outline	Final voice, tone, cultural fit, value-loaded framing	DL
Generating distractors for multiple-choice items		DL

Agent-led tasks	Human-only tasks	Loop
	Construct validity; item-bias review; cut-score setting	
Producing alt-text, transcripts, captions, reading-level adjustments	Confirming UDL fit for the *least-imagined* learner	DL+PL
Running WCAG automated scans, contrast checks, link audits	Adjudicating ambiguous accessibility tradeoffs	DL
Generating Quality Matters self-review evidence packages	Final QM reviewer judgment and rubric application	DL
Pulling and visualizing learning analytics, drop-off, time-on-task, performance metrics	Causal interpretation of *why* learners disengaged or performance moved	PL
Drafting iteration recommendations from outcome data	Deciding which iteration is worth the cost	DL+PL
Producing federal-format deliverable packaging (SCORM, xAPI, 508, classification marking)	CUI/classification review and marking sign-off	DL
Translating between practitioner-facing and reviewer-facing language	Stakeholder trust calls and political reads	DL
Generating evaluation instruments (Kirkpatrick L1–L3 surveys, observation rubrics)	PRS-4 business-impact attribution; performance-causation claims	PL
Synthesizing competitor / prior-art scans for instructional approaches	Strategic positioning of the offering	DL
Drafting facilitator guides and instructor enablement	Live read of room; recovery from learner derailment	DL+PL
Maintaining a living style guide and component library	Deciding when the style itself needs to change	DL
(PRS-3+) Generating per-learner content variations from a validated template	Setting the variation boundaries; auditing for fairness across subgroups	PL
(PRS-3+) Real-time hint and scaffold generation during workflow learning	Defining what counts as success and what counts as derailment	PL
(PRS-3+) Match · Gap · Recommend patterns for performance support	Validating the gap analysis and recommendation logic	PL
(PRS-4) Continuous content adaptation based on learner performance signals	Setting the risk envelope; auditing the adaptation logic	PL

Reading the taxonomy

Most agent-led tasks accelerate the *design loop* (DL): they compress ID cycle time. This is what most published AI-in-ID work focuses on, and the productivity gains are real (the author's federal practitioner observations of first-draft cycle reductions sit here; see §1 and §12). But by itself, design-loop acceleration ships courses faster without proving they move performance.

Performance-loop acceleration (PL) is where the framework's organizing outcome actually plays out. PL tasks cluster at PRS-3+: per-learner variation, real-time scaffolding, performance-support recommendation, continuous adaptation. They belong in the agent column because they are tasks the human ID practitioner *cannot do at scale* (no one can hand-author per-learner content variations for 500 learners) so agent capability on these tasks is genuinely new ground, not substitution.

A concrete, operational instance of PL acceleration in adjacent practice: AR-overlay performance-support tooling of the kind being deployed in industrial-machine maintenance contexts: voice in, expert answer out, automatic logging, layered knowledge across documentation and per-asset history, AR overlay at the moment of work. (Operational deployments are documented in industrial maintenance vendor materials and in trade-press coverage of HoloLens-class deployments; this paper treats the deployment pattern as exemplary rather than as a specific cited case study.) It is not formal training; it is performance support delivered into the workflow, exactly the territory PRS-3+ AAFL implementations occupy. The Apply, Solve, and Change moments of need are served at the point of work, not in a classroom.

The human-only column is the framework's anti-replacement claim. Every entry in it is a decision that requires *learner-in-context judgment*, *value commitment*, *causal reasoning*, *political read*, or *institutional accountability*. None of these is bounded-and-reviewable. None of these compresses with model improvement.

When the line moves

The split is not a permanent claim about AI capability. It is a permanent claim about *what kinds of decisions belong to humans regardless of capability*. As models improve, more rows can move from human-only to agent-led, but only at the production gates (rows 4–7 territory), never at the pedagogical gates (rows 1–3, 8 territory). The pedagogical/production split is the load-bearing distinction; specific row assignments are versioned.

§6. The Proportional Restraint Scale (PRS)

Four PRS levels of HITL implementation. What advances an organization between them is **governance discipline**, not AI capability.

PRS-1 — Assistant

Agent generates suggestions; human authors every artifact. Human reviews 100% of agent output before incorporation. Modest design-loop savings, consistent with the lower end of generative-AI productivity bands measured inside the jagged frontier in adjacent knowledge-work domains (Brynjolfsson et al., 2023; Dell'Acqua et al., 2023). Default starting posture; always appropriate.

PRS-2 — Co-designer

Agent drafts artifacts inside a defined scope (a lesson, an item bank, a storyboard); human edits to final. Human reviews 100% of agent output but no longer authors first-draft prose. Substantial first-draft compression: first-draft cycle reductions in the 40–60% range based on the author's federal practitioner observations across multiple production deliveries (see §1, §1.5 Method, §12 Limitations); this range is consistent with the higher end of generative-AI productivity bands inside the frontier (Dell'Acqua et al., 2023, BCG-Harvard field experiment in knowledge work). Most accredited and federal training programs land here.

PRS-3 — Agent-driven with gates

Agents run production tasks (Source Truth, Equity, Assessment Validity, Release prep) end-to-end inside one phase; humans engage at the eight HITL gates. Output is gate-flagged: clean drafts auto-route through production gates, flagged drafts trigger human review. Per-learner personalization becomes possible: Match · Gap · Recommend territory. First-draft → final compression with gate quality, reallocated to gate quality rather than to volume. **This range is projected, not measured.** AAFL has not yet been operated end-to-end at PRS-3 across a federal program; see §12 (Limitations and Future Work). Performance-loop acceleration becomes real.

PRS-4 — Bounded autonomy

Multi-agent workflows generate per-learner curriculum end-to-end inside an institution-defined risk envelope. Humans spot-audit, set policy, own escalation. Pedagogical gates stay human; production gates become statistical sampling. The framing changes: at PRS-4, you are not "designing a course" anymore, you are running a performance-acceleration system. Continuous performance-loop acceleration in bounded domains. **No measured productivity bands are available at PRS-4.** The published AI-in-ID literature reports no field deployment at this PRS level at this writing; AAFL's PRS-4 prescription is a forward-looking spec, not a measured operating point (see §12 Limitations).

The realistic ceiling

> ***Higher restraint is not better. Appropriate restraint is better.***

Most federal and accredited programs should sit at **PRS-2 or PRS-3**. PRS-4 is the right answer for a narrow set of contexts and the wrong answer for most.

Where PRS-4 is inappropriate:

- **Credentialing and certification programs.** A credential whose content was authored by autonomous agents without human gate-review on each artifact cannot defend itself in audit.
- **Classified content authoring.** No agent autonomy on classified material. Classified content authoring caps at PRS-2 even with approved infrastructure.
- **High-consequence performance training.** Anything where a learner's performance failure has safety, security, or financial consequences requires human-signed production gates.
- **Programs without operational governance discipline.** PRS-4 governance is not a checklist; it is a running system. Organizations that cannot reliably operate PRS-3 governance cannot operate PRS-4 governance at all.

Where PRS-4 is appropriate *(design guidance, not field-deployment claim)*. The list below describes the contexts where a PRS-4 implementation would be defensible by AAFL's own criteria; it is not an assertion that PRS-4 field deployments exist at the time of writing. The PRS-4 prescription is forward-looking; the appropriateness criteria are stated now so that organizations considering a PRS-4 design have the framework's view on where the design would and would not fit.

- **High-volume skill refreshers** (annual compliance refresh for a 10,000-person workforce)
- **Onboarding pathways** (role-based onboarding where the content space is bounded)
- **Performance support continuum** (real-time recommendations within a defined competency space: the Match · Gap · Recommend pattern, or AR-overlay performance-support tooling deployed at the point of work in industrial-machine maintenance contexts)
- **Adaptive remediation** (agent-driven re-teaching loops *after* the learner has failed a human-validated assessment)

Performance and what the framework actually delivers

The PRS is not just "how much can the agent do." It is **how much workplace performance acceleration the system delivers**:

Level	Design-loop impact	Performance-loop impact
PRS-1	ID-design speed gains only	None
PRS-2	Training-development cycle compression (40–60% first-draft, practitioner observation; see §1, §12)	Limited — same content for all learners, produced faster
PRS-3	First-draft → final compression with gate quality	

Level	Design-loop impact	Performance-loop impact
		Real — per-learner personalization; performance-support generation
PRS-4	"Designing a course" reframes as "running a performance-acceleration system"	Continuous — real-time learner-to-content fit in bounded domains

PRS-1 / PRS-2 are *productivity* stories (faster ID work). PRS-3 / PRS-4 are *capability* stories (work that wasn't possible before). The performance-acceleration claim only becomes credible at PRS-3+.

How to choose the right level

Three questions, asked in order:

1. **What is the consequence space?** The consequence space sets the PRS ceiling:

Consequence space	PRS ceiling
Credentialing / classified / safety-critical	Cap at **PRS-2**
CUI / accredited but not classified	Consider **PRS-3**
Public-trust unclassified high-volume	Consider **PRS-4**

2. **What governance is operationally running?** PRS-1 governance means a working audit trail (a record of which agent ran which task with which model against which source, with the human gate decision logged); reliable handling of agent-output classification; and a documented escalation path when the human reviewer disagrees with the agent. If the organization cannot reliably operate that PRS-1 baseline, do not aspire to PRS-3.

   > ***PRS readiness follows operational capacity, not desire.***

3. **What is the performance claim?** If the goal is "ship courses faster," PRS-2 suffices. If the goal is "compress time-to-competency for a 10,000-person workforce," PRS-3+ becomes the answer. Match the level to the claim.

If any answer is unclear, default to one level lower than the question would suggest. *Higher restraint is not better; appropriate restraint is better.*

How the PRS relates to adjacent autonomy and risk scales

The PRS is not the only scale in the field that grades human-AI work along an autonomy axis. Three adjacent references deserve explicit positioning, because reviewers will reach for them and ask why AAFL did not simply adopt one.

SAE J3016 (driving automation levels 0–5) (SAE International, 2021). The SAE on-road automation standard grades vehicle systems from no automation (L0) through full driving automation (L5) by reference to who is responsible for the dynamic driving task and under what conditions. The PRS borrows the *prescribed-ceiling-by-context* shape (different operational design domains permit different levels) but the substantive content is incomparable. SAE J3016 grades a single closed-loop control task; the PRS grades a portfolio of pedagogical and production tasks where the human-only commitments at the pedagogical gates do not relax at any level. PRS-4 is not "L5 for instructional design"; even at PRS-4, four pedagogical gates remain humans-only. The shape is shared; the autonomy ladder is not.

NIST AI Risk Management Framework (AI RMF 1.0) (National Institute of Standards and Technology, 2023). NIST's AI RMF is a *risk-governance* framework: its Govern, Map, Measure, Manage functions describe how an organization characterizes and manages AI risk across a system's lifecycle. AAFL's Governance Layer (§8) and the PRS-caps-by-classification framing (§10) are intentionally compatible with the AI RMF: an organization operating AAFL at PRS-2 or PRS-3 on federal training work should be able to map its audit trail, classification handling, and escalation paths onto the AI RMF's Manage function without rework. The relationship is layered, not competing: NIST AI RMF tells an organization *how to govern AI risk in general*; AAFL tells an instructional-design practice *which autonomy ceiling applies to which training context, and which gates produce the artifacts the AI RMF's Manage function consumes*. AAFL is not a substitute for AI RMF compliance; it is the ID-specific operating layer on top.

Mollick's autonomy gradient / centaur-cyborg distinction (Mollick, 2024; Dell'Acqua et al., 2023). Ethan Mollick has popularized a working distinction between *centaur* use of AI (clear division of labor between human and agent, human owns the integrated output) and *cyborg* use (deeply interleaved, agent operating inside the human's working process). The PRS is consistent with this distinction but operates at a different grain: PRS-1 and PRS-2 are centaur configurations under Mollick's terminology; PRS-3 and PRS-4 begin to blend toward cyborg configurations in the production-gate territory while remaining centaur at the pedagogical gates. AAFL's contribution is not the gradient itself (Mollick's framing is the cleaner generic articulation) but the *prescription* of which gradient point fits which instructional-design context with which classification of data.

The PRS, in short, is not redundant with these scales; it is the instructional-design-specific operating ladder that consumes from the general-purpose risk governance (NIST), borrows the prescribed-ceiling shape from operational-domain regulation (SAE), and instantiates the centaur-cyborg gradient (Mollick) at gate-level granularity. Treating the PRS as the only scale would be a category error; ignoring the adjacent scales when operating the PRS in federally-regulated contexts would be a compliance error.

§7. The Eval Framework

How to evaluate an AI-designed instructional intervention. Borrows the eval-driven discipline from agent engineering practice ("*define evals before agents fulfill them*") and anchors it in workplace performance as the organizing outcome.

The discipline

> *The eval rubric for a given course is authored at the* ***Performance Outcome Gate*** *(HITL Gate 2), before any Develop-phase agent runs. Agents are then evaluated against those criteria.*

This is the single biggest defense against the agent-era failure mode of *artifacts that are well-formed but not instructionally sound*. An agent that doesn't know what success looks like will produce confident, professional, fluent output that misses the actual outcome, and a reviewer who is checking surface fluency will let it through. The Evaluation-as-Spec Layer makes this discipline **blocking**: neither the Design nor the Develop phase can run until the eval rubric is signed at Gate 2.

Six dimensions, ordered by priority

The six dimensions are not drawn from a single tradition. *Performance attainment* and *outcome attainment* sit in the HPT and Kirkpatrick lineage (Gilbert, 1978; Brinkerhoff, 2003; Kirkpatrick & Kirkpatrick, 2016) supplemented by the systematic-design tradition's criterion-referenced framing (Dick, Carey, & Carey, 2014; Mager, 1962). *Construct validity* is the psychometric tradition, drawing on Messick's (1989) modern construct-validity framework. *Pedagogical fidelity* is the program-evaluation and implementation-fidelity literature (Dane & Schneider, 1998; O'Donnell, 2008). *Learner experience* draws on cognitive-load theory (Sweller, 2011), Mayer's (2014) cognitive theory of multimedia learning extending Sweller into the multimedia-design space the agent era operates in, the cognitive-architecture grounding of learning-for-instruction (Driscoll, 2005), and the LX/UX adaptation of HCI methods to learning artifacts. *Production integrity* is the federal-compliance and software-QA tradition (WCAG, Section 508, FedRAMP, NIST 800-171). The dimensions are organized as a single rubric not because they share an epistemology but because they share a *gate*: every artifact at Gate 6 (Assessment Validity), Gate 7 (Release), and Gate 8 (Performance-Effect) is judged against all six. Mixing traditions is the explicit choice; pretending the mix is monolithic would be the methodological error.

1. Performance attainment (primary spine)

The non-negotiable. Did workplace performance actually accelerate?

Measures: time-to-competency; error rate change; throughput; transfer-to-job; retention over time (30/60/90-day).

Why primary. Workplace performance is the framework's organizing outcome. An intervention that does not move performance is a learning experience, not a performance-acceleration intervention.

The framework is opinionated: this dimension is the spine, not one of six equal dimensions. This is Kirkpatrick L3 (Behavior) and L4 (Results) territory.

Failure mode this catches. Beautiful, well-formed courses that learners complete and pass and that change nothing about workplace performance. The phrasing can read as a contradiction (*how can a well-formed course change nothing?*) but the contradiction is the point. A well-formed course is a *necessary* condition for performance gain, not a *sufficient* one. The conceptual sharpening from Bender et al. (2021) applies here: well-formed text that is statistically plausible but indifferent to whether the underlying claim is right is the agent-era specialization of the "stochastic parrot" failure mode; AAFL's pedagogical gates are the structural answer to the question Bender et al. pose: *who decides whether the model's output is the kind of object it appears to be?* Transfer-to-job further depends on opportunity to practice, reinforcement structures, manager behavior, performance-support availability at the moment of work, and the gap between training-environment fidelity and on-the-job conditions. The course can satisfy every learning-attainment criterion and still fail Kirkpatrick L3/L4 because it never made contact with those downstream conditions. This is exactly why AAFL anchors in workplace performance rather than course completion: the framework refuses to let well-formedness substitute for outcome.

2. Outcome attainment

Did learners demonstrate the stated objectives at the stated mastery threshold? Pre/post, transfer task, performance observation. Kirkpatrick L2.

A learner can pass and not transfer (false positive on the assessment). A learner can fail the assessment and still improve workplace performance (false negative on the assessment). Both dimensions matter; #1 is primary because it is the outcome the institution is buying.

3. Construct validity

Does the assessment actually measure the construct claimed? Item analysis, expert review, fairness audit.

The agent-generated assessment is the place where surface plausibility is most easily mistaken for instructional soundness. Items can be well-formed, distractors plausible, and the whole assessment can measure a subtly different construct than the one stated. Construct validity review at HITL Gate 6 is what catches this.

4. Pedagogical fidelity

Did the implementation match the design intent? Detects drift between Gate 3 (Pedagogical Strategy) decisions and the artifact actually produced; between Gate 2 (Performance Outcome) criteria and assessment items at Gate 6; between learner persona at Gate 1 and content reading level / cultural fit / contextual examples.

Agents drift between phases — this is a *practitioner observation* in AAFL implementation work, not a claim grounded in a specific peer-reviewed study of multi-agent ID workflows. The closest empirical adjacencies are the LLM literature on long-context degradation and lost-in-the-middle effects (Liu et al., 2024) and the jagged-frontier work on uneven AI capability across adjacent tasks (Brynjolfsson et al., 2023; Dell'Acqua et al., 2023), which together establish that *anchoring* and *capability* both vary across sequential agent invocations even when the brief looks stable. The application to phase-to-phase ID workflows extrapolates from these adjacencies rather than measuring the phenomenon directly. Fidelity checks at Gate 6 close the loop in practice; whether they would close it under controlled study of AAFL-shaped workflows is a Future Work item (§12).

5. Learner experience

The learner-side construct. Did the intervention work for the learner across the population, not just the median case?

Measures: cognitive load fit (Sweller-tradition working-memory considerations applied to the artifact's information density and sequence; Mayer-tradition multimedia-design principles applied to modality choice); engagement (drop-off rates, replay rates, voluntary continuation past required completion); equity-of-experience across subgroups (does the median learner succeed while the tails are abandoned? are there subgroup-by-outcome interactions worth surfacing?).

Why distinct from #2 (Outcome attainment). A learner can meet the objective and have had a bad experience: high cognitive load, low engagement, equity gaps in supporting subgroups. Learner Experience and Production Integrity (the next dimension) are kept separate because they answer fundamentally different questions: *did the learner have the experience the design intended* (this dimension) vs *was the artifact infrastructurally fit to ship* (the next dimension). Conflating them obscures the trade-offs.

6. Production integrity

The infrastructure-side construct. Was the artifact infrastructurally fit to ship?

Measures: source-traceability (every claim citable to a validated source; no hallucinated authority); accessibility conformance (WCAG 2.1 AA, Section 508, alt-text completeness, transcript accuracy, contrast compliance); classification compliance (CUI marking, FedRAMP, NIST 800-171 where applicable; ITAR/EAR where applicable).

Why distinct from #5 (Learner experience). Production integrity dimensions are *runnable as checks against spec* (accessibility scanners, citation auditors, classification-marking validators) without a learner in the loop. Learner-experience dimensions require learner data. In federal contexts, production integrity is gating, not advisory: a CUI training program that ships without classification compliance (for example, source material that should have been marked CUI but was reproduced unmarked into a learner-facing module, or an artifact stored in a tenant cleared only for FOUO) is not primarily an instructional design problem. It is a federal compliance incident. The ID team owns the consequences (rework, retraction, possible reporting obligation), the program owns the audit-finding exposure, and the institution owns the accreditation risk. AAFL treats production integrity as gating to keep this category of failure out of the path to release in the first place.

Eval framework × Proportional Restraint Scale

The rubric is authored at Gate 2 before Design at every PRS level: its existence is constant. What changes across PRS levels is *what runs against it* and *how much is reviewed*:

- **PRS-1 and PRS-2.** Human or human-led drafts are evaluated against the rubric; 100% human review of every artifact.
- **PRS-3.** Agents run production tasks against the rubric; humans review agent-flagged outputs plus a statistical sample of the rest.
- **PRS-4.** Continuous, real-time evaluation of agent outputs against the rubric; humans audit aggregate eval performance rather than individual artifacts.

Refer back to §6 for the substantive PRS level definitions; this section only restates the eval-discipline implications at each level.

What the eval framework requires

The framework refuses to let Develop run until the practitioner can answer:

1. What workplace performance metric will move (and how will we measure it)?
2. What is the mastery threshold for the stated objectives?
3. What construct does each assessment item measure, and how will we know?
4. What pedagogical strategy has been committed, and what is fidelity drift?
5. What does equity-of-experience look like for this learner population?

If the practitioner cannot answer these before Design, the rubric is not yet authored.

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§8. The Three Cross-Cutting Layers

Architectural depth on the three layers that did not exist in classic ADDIE.

Orchestration

Routes work between agents and humans. Decides:

- **Which agent runs which task.** A specialized item-writer agent handles distractor generation; a different agent handles facilitator-guide drafting; orchestration knows which is appropriate.
- **Which model is appropriate.** Frontier models for high-judgment drafting; cost-optimized models for high-volume formatting; government-tenant generative AI tools (only environments cleared for the relevant data classification) for CUI work. Orchestration is the layer that enforces classification-based model selection.
- **When artifacts route to which gate.** A drafted assessment item routes to Gate 6 (Assessment Validity) for human review at PRS-1 / PRS-2; at PRS-3+, agent-confidence-flagged items route to Gate 6 while clean items route directly to Validate.
- **What gets logged.** Every agent invocation, every model version, every human gate decision. The orchestration log is the input to the Governance Layer audit trail.

The technical-substrate articulations of orchestrator-workers patterns in agent-engineering practice (Anthropic Research, 2024) are a useful runtime reference for how this layer is built; AAFL operates at the design-discipline level above the runtime substrate.

Evaluation-as-Spec

Evaluation-as-Spec is a practice borrowed from agent engineering (Anthropic Engineering, 2026) where the criteria an agent's output will be judged against are defined as a structured artifact (the *eval rubric*) *before* the agent runs. The rubric is the spec the agent fulfills. The discipline inverts the conventional Design-then-evaluate flow: instead of building first and checking against criteria afterward, AAFL requires the criteria to exist, in writing, before any building begins. This is the framework's structural answer to the agent-era failure mode of *artifacts that are well-formed but not instructionally sound*.

The blocking discipline. Neither the Design nor the Develop phase can run until:

1. The Performance Outcome Gate (Gate 2) has been signed
2. The eval rubric is authored, including all six dimensions (Performance attainment, Outcome attainment, Construct validity, Pedagogical fidelity, Learner experience, Production integrity)
3. The rubric defines measurable criteria, not aspirations

Without a blocking eval rubric, agents will produce surface-fluent content that drifts from the intended outcome. The Layer makes the rubric a precondition, not a postcondition.

Governance

Institutional accountability. Captures:

- **Audit trail.** Chain of custody for every agent-generated artifact: model version, prompt-and-response, human reviewer signature, gate-flag history, classification marking. Retained for the contract retention window.
- **Classification handling.** FedRAMP, NIST 800-171, CUI marking, Section 508 conformance, ITAR/EAR where applicable.
- **Accreditation alignment.** ACE evaluator expectations; Quality Matters Standards 2 (Learning Objectives), 3 (Assessment), 8 (Accessibility); WCAG 2.1 AA + Section 508 conformance evidence.
- **Escalation paths.** Defined human-side decision protocol for when the human reviewer disagrees with the agent's output, when human override is logged, when the disagreement escalates to a senior reviewer or governance board.
- **PRS caps by classification.** No PRS-4 on classified content. PRS-3 only on approved federal-tenant infrastructure. PRS-4 only on public-trust unclassified high-volume work. Caps are *enforced* by the Layer, not aspirational.

The federal claim AAFL can defensibly make: among the published AI-era ID frameworks reviewed in this work (see Appendix B for the comparative scan: ADDIE, FRAME™, Anthropic 4Ds, Chai et al., EduPlanner, AUSS, Action Mapping), AAFL appears to be the only framework that prescribes PRS caps tied to data classification. That is exactly the question federal contracting officers ask first.

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§9. Engagement With Prior Frameworks and Intellectual Lineage

AAFL enters the literature as a coherent extension and integration of existing work, not as a clean-break replacement. The engagement here is substantive and structural rather than plank-by-plank: AAFL inherits a tradition of thought rather than borrowing isolated techniques. The reader should be able to recognize the lineage without being able to reduce any one feature of the architecture to any single source.

The performance tradition

The Human Performance Technology (HPT) tradition is the bedrock AAFL inherits: a body of work that has insisted on workplace performance as the legitimate outcome of L&D investment for more than half a century. The decisive intellectual contributions in that tradition include Thomas Gilbert's *Human Competence* (1978), which reframed the unit of analysis from the learner's behavior to the performer's output and gave the field its enduring distinction between behavior and accomplishment; Geary Rummler and Alan Brache's *Improving Performance* (1990), which treated workplace performance as a systems-level property; and the long-running professional infrastructure of the International Society for Performance Improvement, which sustained the discipline through periods when the rest of L&D was preoccupied with course completion as the win condition.

Workplace-oriented operational articulations of the same commitment include Cathy Moore's Action Mapping, introduced on her practitioner blog c. 2008 and consolidated in *Map It* (Moore, 2017), which insists on backward design from business performance goals; Conrad Gottfredson and Bob Mosher's 5 Moments of Need (Gottfredson & Mosher, 2011), which draws the line between formal learning (the New and More moments) and performance support (Apply, Solve, Change) and locates the bulk of workplace skill acquisition outside formal training; Allison Rossett and Lisa Schafer's performance-support taxonomy of Planners, Advisors, Coaches, and Sidekicks (Rossett & Schafer, 2007), which named the categories of in-workflow help that AR-overlay tooling now operates; and Robert Brinkerhoff's Success Case Method (Brinkerhoff, 2003), which provided the methodology for tracing whether an intervention actually moved performance for the workers who used it.

The Kirkpatrick L1–L4 vocabulary (Kirkpatrick & Kirkpatrick, 2016; originally articulated by Donald Kirkpatrick in 1959) is the dialect every reviewer of AAFL will use; AAFL adopts the vocabulary natively *while engaging the substantive critique of it that the L&D research community has accumulated over four decades*. Kraiger, Ford, and Salas (1993) argued that the four-level model conflates evaluation taxonomy with causal model: moving up the levels was never empirically a chain of causation, and treating it as one license-poor inference. Holton (1996) made the case more sharply: the model is a *taxonomy of outcomes*, not an evaluation *theory*, and using it as a theory has produced decades of weak transfer claims. AAFL inherits this critique and responds to it structurally rather than terminologically. The framework retains L3/L4 *language* because it is the operational vocabulary federal reviewers, ACE evaluators, and corporate L&D leaders use, but it does not assume that Gate 8 evidence of L3 behavior change automatically implies L4 results; nor does it assume that L1 satisfaction or L2 learning predicts L3 transfer. The transfer-of-training literature (Baldwin & Ford, 1988; Burke & Hutchins, 2007) is the deeper anchor: transfer is governed by trainee characteristics, training design, and work environment, and the work-environment variable is exactly what AAFL's Gate 8 *while-you-work* signal is designed to surface. Where Kirkpatrick gives AAFL the vocabulary to be read by the field, Baldwin-Ford and the broader transfer-of-training tradition give it the causal vocabulary to do the actual analytical work.

These works do not appear in AAFL as plank-by-plank derivations. They appear as bedrock: the assumption that workplace performance is the legitimate organizing outcome, that course completion is a weak proxy, that performance support belongs in the workflow rather than in the classroom, and that the eval question is *did performance move*, not *did the learner pass*. AAFL does not paraphrase any one of these voices; it operates inside the consensus they built.

ADDIE as the architectural heritage

ADDIE (Branson et al., 1975, FSU). AAFL is ADDIE practiced at agent speed with HITL gates and three new layers. *Continuity, not replacement.* ADDIE Guide remains the reference; AAFL is the operating manual. The decision to preserve ADDIE's vocabulary is the framework's single largest concession to its audience: federal contracting officers, ACE evaluators, Quality Matters reviewers, and military training command staff all read in ADDIE's vocabulary, and a framework that begins by replacing the vocabulary forfeits the audience most likely to adopt it.

Iterative design heritage

SAM (Allen & Sites, 2012). AAFL inherits SAM's iterative discipline. The Translator's Loop is functionally similar to SAM's rapid-prototyping cycle, applied at per-artifact granularity rather than per-course granularity.

Pedagogical-design heritage

AAFL's pedagogical gates (Learner Reality, Performance Outcome, Pedagogical Strategy, and Performance-Effect) do not invent the underlying instructional-design theory. They operate against a body of work that any ID reviewer will expect to see named.

Gagné's *Conditions of Learning and Theory of Instruction* (Gagné, 1985; 4th ed., original 1965) provides the cognitive-architecture grounding for the conditions under which different categories of learning outcome can be reliably produced. AAFL's Gate 3 (Pedagogical Strategy) is the framework's structural commitment to asking *which conditions does this strategy create for this learner population*; the question is Gagné's even when the answer is agent-drafted. **Dick, Carey, & Carey's *Systematic Design of Instruction*** (Dick et al., 2014; 8th ed., original 1978) is the canonical systematic-design textbook against which any ADDIE-extension framework will be read; AAFL inherits its commitment to backward-mapped objectives and criterion-referenced assessment, both operationalized at Gate 2 (Performance Outcome) and Gate 6 (Assessment Validity). Smith and Ragan's *Instructional Design* (Smith & Ragan, 2005) is the parallel ID-textbook anchor that any reviewer trained outside the

Dick-Carey tradition will read AAFL against; the framework's commitments are consistent with both lineages.

Merrill's First Principles of Instruction (Merrill, 2002), published in *Educational Technology Research and Development*, consolidates the cross-theoretic instructional principles (task-centeredness, activation of prior knowledge, demonstration, application, integration) that AAFL's pedagogical gates assume without restating. A practitioner who passes Gate 3 with a strategy that violates Merrill's principles has produced a pedagogically thin design; the framework does not relitigate Merrill, it operates on top of him.

Reigeluth's *Instructional-Design Theories and Models* series (Reigeluth, 1999, and subsequent volumes co-edited with Carr-Chellman in 2009 and with Beatty and Myers in 2017) catalogs the theoretical pluralism within which AAFL is one operating choice. AAFL takes an explicitly *performance-acceleration-oriented* stance within that pluralism; it does not claim that workplace performance is the only legitimate organizing outcome of L&D, only that it is the legitimate organizing outcome for the workplace contexts AAFL addresses (see §12 Limitations on K-12 scope).

Universal Design for Learning (Meyer, Rose, & Gordon, 2014; CAST, 2018) is the theoretical anchor underneath AAFL's Gate 5 (Equity & Accessibility). The framework's operational instance (automated WCAG 2.1 AA conformance scans, alt-text, transcripts, captions, reading-level analysis) is the production-gate layer; the substantive commitment is UDL's *multiple means of engagement, representation, and action/expression* applied to the agent-era authoring stack. Gate 5's human criterion ("Does this work for the learner whose context I have least imagined?") is UDL's commitment translated into a single reviewer-facing question.

These anchors are named here, not at individual gates, in keeping with the lineage discipline established in §9's opening paragraph: AAFL is a synthesis, not a composite of borrowed parts.

Scaffolding as the underlying construct

AAFL's pedagogical gates are, in effect, *scaffolding contracts* for the human-agent pair. The Vygotskyan Zone of Proximal Development (Vygotsky, 1978) and the scaffolding tradition it grounds (Wood, Bruner, & Ross, 1976) provide the underlying construct: the practitioner's gate judgments are the scaffolds that hold the agent's output inside the zone of pedagogically defensible work, just as a teacher's scaffolds hold a learner's developing performance inside the zone of competence the learner is reaching for. The framework does not relitigate scaffolding theory; it operates inside it. Naming the construct explicitly here matters because reviewers from the learning sciences tradition will read AAFL as a scaffolding framework whether the paper says so or not; saying so makes the inheritance auditable.

Agent-era frameworks

FRAME™ (Hardman, 2025). FRAME maps capability risk; AAFL maps decision authority. FRAME tells you *which AI to trust for which task*; AAFL tells you *which task humans must keep regardless of which AI you trust.* The frameworks are complementary, not competing; citing FRAME strengthens

AAFL's positioning. Hardman's "danger zone" framing in the jagged-frontier work (Hardman, 2025) is the closest published precursor to AAFL's pedagogical-vs-production task split.

Anthropic AI Fluency Framework / 4Ds (Dakan & Feller, 2023–2025). The 4Ds (Delegation, Description, Discernment, Diligence) are practitioner *competencies*; AAFL is the *process* those competencies operate within. Discernment lives at every HITL gate; Delegation defines the agent-vs-human task split; Description briefs the agent at every gate; Diligence is the audit trail the Governance Layer records. *Disclosure:* the author completed Anthropic Academy's *AI Fluency for Educators* and *AI Fluency: Framework & Foundations* courses as a learner; AAFL is informed by but independent of those curricula, and the author has no instructional or advisory role at Anthropic.

Chai et al. (2025). Peer-reviewed phase-by-phase synthesis of generative AI in instructional system design, published in *Human Resource Development Review*. The academic anchor for the position that AI integration is phase-specific, not monolithic. AAFL extends Chai et al. by adding the gate taxonomy, the PRS with caps, the Evaluation-as-Spec Layer, and workplace performance as the eval spine.

Multi-agent academic systems: EduPlanner (Zhang et al., 2025) and AUSS (Arya Mary et al., 2026). Cited as prior art on multi-agent ID systems. AAFL distinguishes itself by foregrounding *human gates*, not multi-agent collaboration. EduPlanner's Evaluator/Optimizer/Question-Analyst is an agent harness; AAFL's eight gates are human criteria. The AUSS architecture similarly operates as an inter-agent collaboration system across student, educator, and institutional levels; AAFL takes no position on multi-agent collaboration as a technical strategy, but holds that the *human gate* is the unit of work that institutional accountability flows through.

Methodological foundations

Eval-driven development (Anthropic Engineering, 2026; Liang et al., 2023; EleutherAI, 2024). Source of the technical discipline borrowed for the Evaluation-as-Spec Layer. The discipline is vendor-neutral and well-established in the broader LLM-evaluation literature; Anthropic Engineering's articulation is the cleanest practitioner-facing version cited here, and Stanford's HELM benchmark (Liang et al., 2023) and EleutherAI's lm-evaluation-harness (EleutherAI, 2024) are named to make the non-vendor lineage visible. AAFL's eval-as-spec stance does not depend on any specific vendor's tooling.

Centaur models (Saghafian & Idan, 2024). Hybrid human-algorithm symbiosis pattern, articulated in *Harvard Data Science Review*. The Translator's Loop is functionally a centaur cycle applied to instructional artifact production.

Human + Machine (Wilson & Daugherty, 2018, updated 2024). The augment-not-replace operating principle, articulated for organizational practice generally. AAFL localizes this thesis to ID specifically.

Empirical AIED baseline (VanLehn, 2011). The most-cited meta-analytic anchor in AIED for "where does AI tutoring sit against human baseline." VanLehn (2011) found intelligent tutoring systems with step-based or sub-step interaction granularity approached the effect-size of human one-on-one tutoring, refining a generation of overstated claims about the *2-sigma* gap between classroom and tutoring (Bloom, 1984). AAFL's PRS-3+ performance-loop claims (per-learner variation, real-time scaffolding, performance-support generation) sit in the territory VanLehn's meta-analysis characterized; the framework is the operating discipline that lets institutions build *toward* that effect-size in instrumented contexts without claiming the LLM-era tooling has already achieved it.

Brynjolfsson, Li, & Raymond, *Generative AI at Work* (2023). NBER Working Paper 31161, subsequently published in the *Quarterly Journal of Economics* (Brynjolfsson et al., 2025): the most-cited empirical study of generative-AI productivity effects in a customer-support setting, establishing the productivity-impact range that the §6 PRS-1/PRS-2 bands are anchored against. **Dell'Acqua et al.** (2023, BCG-Harvard field experiment, HBS Working Paper 24-013, subsequently published in *Organization Science*, 2026) is the companion knowledge-worker field-experimental study cited in §6 as the productivity-band anchor for the higher end of the PRS-2 range, and the canonical source of the "jagged frontier" construct that organizes AAFL's task-by-task capability-uneven framing.

Engagement with the critic camp

A framework that proposes to operationalize agent capability in instructional design owes the reader a substantive engagement with the critical AI-in-education tradition, not just the productivity-oriented and HPT-oriented traditions. Several concerns from that tradition shape AAFL's design choices even when they are not named at individual features.

Selwyn's *Should Robots Replace Teachers?* (Selwyn, 2019) raises the deskilling concern: that delegating authorship to AI agents hollows out the practitioner profession over time, leaving humans only the credentialing-and-rubber-stamping residue of work the agent now drafts. AAFL's structural response is the four pedagogical gates: Learner Reality, Performance Outcome, Pedagogical Strategy, and Performance-Effect are *humans-only at every PRS level*, including PRS-4. The framework's anti-replacement claim is not a sentimentality about human craft; it is a structural commitment that the value-loaded decisions of instructional design do not relax with model capability. The deskilling concern is taken seriously precisely by refusing to let production-gate automation creep across the pedagogical line. AAFL takes no position on the broader political economy of edtech procurement Selwyn examines — vendor selection, lock-in risk, and institutional pressure to adopt are governance questions the Governance Layer (§8) recognizes but does not resolve.

Williamson's *Big Data in Education* (Williamson, 2017) raises the surveillance and data-political concerns inherent in learner analytics, adaptive systems, and per-learner content variation: exactly the territory PRS-3 and PRS-4 operate in. AAFL's Governance Layer (§8) and classification-aware caps (§10) are partly a structural response: audit trail, classification handling, escalation paths, and named-accountable-human signatures are the institutional infrastructure that prevents AAFL implementations from becoming the surveillance-and-optimization regimes Williamson describes. Williamson's deeper concern — that datafication itself reshapes what counts as learning — is not fully resolvable inside any operating framework. AAFL's narrower response is that workplace-performance signals (time-to-competency, error-rate change, transfer-to-job) are *causal* claims about what an intervention changed, not *representational* claims about who the learner is; that distinction is what allows the framework to operate in measurement-instrumented contexts without claiming

Williamson's concern is dissolved. PRS-4 is *bounded* autonomy, bounded by classification, by domain, by named risk envelope, not unbounded.

Holmes, Bialik, & Fadel's *Artificial Intelligence in Education* (Holmes et al., 2019) is the most-cited recent overview of AI's promises and implications for teaching and learning; AAFL inherits its commitment to taking both the affordances and the risks seriously, and its emphasis on educator agency in the face of vendor-driven adoption pressure. Holmes et al. offer a tripartite distinction between learning *with* AI, learning *about* AI, and learning *despite* AI. AAFL is, by design, a framework for the first: it operationalizes AI as an instrumental aid to instructional design. The second (AI literacy as curriculum content) is adjacent to AAFL but not the framework's claim space; the third (AI as a condition educators and learners must work around) is taken seriously through the Governance Layer and the PRS caps but is acknowledged as not fully addressable by any operating framework. The framework's *cite-don't-pitch* engagement principle is partially indebted to Holmes et al.'s posture of substantive engagement rather than evangelism or rejection.

Luckin's *Machine Learning and Human Intelligence* (Luckin, 2018) is the most-cited contemporary voice in the AIED-and-pedagogy intersection from the European center of this conversation. Luckin's distinction between *artificial intelligence* and *intelligence augmentation* — and her seven-element model of human intelligence (academic, social, meta-cognitive, meta-subjective, meta-contextual, accurate self-efficacy, perceived self-efficacy) — sits directly alongside AAFL's augmentation stance but with a different theoretical anchor. AAFL's pedagogical-vs-production task split is consistent with Luckin's argument that human intelligence is multifaceted in ways the agent does not (yet) match; the production gates are agent-permitted because they sit closer to the bounded sub-elements of intelligence Luckin describes, and the pedagogical gates are humans-only because they sit closer to the meta-elements (meta-cognitive, meta-subjective, meta-contextual) where the agent's structural deficits are most pronounced.

Hicks, Humphries, & Slater's "ChatGPT is bullshit" (2024, *Ethics and Information Technology*) sharpens the conceptual framing AAFL's Source Truth gate (Gate 4) inherits: LLM output is better understood as Frankfurt-style bullshit (indifference to truth) than as hallucination (a perceptual failure the model has). The vocabulary matters because *hallucination* implies the model could in principle correct itself with better data or scaffolding, while *bullshit* implies the model is structurally indifferent to whether its output is true and will remain so regardless of capability gains. Gate 4 is the structural answer either way: the human practitioner's truth-tracking commitments are what the agent does not have.

Watters's *Teaching Machines* (Watters, 2021) provides the historical context the contemporary AI-in-education discourse often omits: the dream of personalized algorithmic instruction is a century old, has been repeatedly oversold, and has repeatedly produced unintended consequences. AAFL's "*PRS-4 is the right answer for a narrow set of contexts and the wrong answer for most*" stance (§6) is exactly the conclusion Watters's history would predict; the framework's posture of opinionated restraint is partially indebted to the corrective Watters's historical work provides against the field's recurring overpromise.

The framework does not claim to resolve the critic camp's concerns; those concerns are about the political economy of AI-in-education and the institutional cultures that adopt these tools, not about framework architecture. AAFL's claim is narrower: the structural choices the framework makes (pedagogical gates as humans-only, caps by classification, blocking eval discipline, audit-trail governance) are the framework architecture's contribution to a problem space the critic camp has correctly named.

What AAFL adds that the field does not yet integrate

Across all of the cited work, no published framework integrates: an HITL decision-gate taxonomy with explicit pedagogical-vs-production classification; a Proportional Restraint Scale (PRS) with prescribed caps by data classification; a performance-anchored eval framework that *blocks* the Develop phase; and a governance layer that enforces the caps rather than describing them. AAFL's contribution is the integration, not any individual element.

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§10. Federal Applicability

For classified, CUI, and federally-accredited contexts, three things change in AAFL.

The Source Truth, Release, and Production Integrity gates become legally non-negotiable

Audit trail must satisfy FedRAMP, NIST 800-171, and contract-specific marking requirements. The Governance Layer captures chain of custody for every agent-generated artifact, model version recorded, prompt-and-response archived for the contract retention window. Section 508 conformance evidence is retained alongside the artifact-of-record.

PRS caps by classification

- **Classified content authoring caps at PRS-2.** No agent autonomy on classified material. Classified content authoring is human-led with agent assistance only.
- **CUI training development can reach PRS-3 with approved infrastructure.** Government-tenant generative AI tools only, in environments cleared for the relevant data classification. PRS-3 is *not* available on commercial-tenant frontier models for CUI work.
- **Public-trust and unclassified high-volume work can reach PRS-4.** Annual compliance refreshers, role-based onboarding pathways, performance-support adaptive remediation in bounded domains.

AAFL prescribes the cap by classification, not by capability. This is the framework's distinctive federal claim.

Accreditation alignment

The eight gates map cleanly to accreditation expectations:

- **ACE evaluator expectations.** Gate 2 (Performance Outcome) and Gate 6 (Assessment Validity) directly address ACE's competency-based evaluation criteria.
- **Quality Matters Standards.** Gate 2 addresses QM Standard 2 (Learning Objectives); Gate 6 addresses QM Standard 3 (Assessment); Gate 5 addresses QM Standard 8 (Accessibility). QM is named here because it is the rubric most frequently cited in the federal and higher-education contexts AAFL addresses; equivalent mappings to OLC OSCQR, iNACOL, and ACCJC distance-education standards are possible and intended in future revisions, and AAFL's gate structure is rubric-agnostic by design.
- **WCAG 2.1 AA + Section 508.** Gate 5 (Equity & Accessibility) is the conformance gate; Section 508 evidence is part of the Release Gate (Gate 7) artifact-of-record.
- **UDL fit.** Checked at Gate 3 (Pedagogical Strategy); validated at Gate 5.

The framework's documentation artifact-of-record is *designed* to drop into an ACE submission packet or a Quality Matters self-review evidence package.

The federal-applicability claim

Among the published AI-era ID frameworks reviewed in this work (ADDIE, FRAME™, Anthropic 4Ds, Chai et al., EduPlanner, AUSS, and the HPT performance-orientation lineage; see Appendix B for the comparative scan), AAFL appears to be the only framework that prescribes PRS caps tied to data classification. This is the question federal contracting officers ask first when evaluating AI integration proposals. FRAME™, the 4Ds, and Chai et al. are all classification-agnostic. AAFL's classification-aware caps are not a restriction; they are the source of its federal credibility.

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§11. Risks and Pitfalls

Six real risks, named so they can be defended. *This section and §12 (Limitations and Future Work) sit back-to-back deliberately and answer different questions. §11 names operational risks: what can go wrong when AAFL is implemented. §12 names epistemic limits: what AAFL has not yet proven about itself. A risk is a thing the framework's operators must mitigate; a limitation is a thing the framework's authors and reviewers must address. Both are published to keep AAFL honest about what it is and what it is not.*

Risk 1 — PRS drift

Organizations read the model as a ladder and push past their appropriate level.

Mitigation. Explicit "realistic ceiling" language in §6; case studies of PRS-2 and PRS-3 wins, not just PRS-4; the "default to one level lower than the question would suggest" rule.

Risk 2 — Eval theatre

Teams adopt the eval framework as documentation rather than as a constraint on the agent loop.

Mitigation. The Evaluation-as-Spec Layer is *blocking*: agents cannot run Design or Develop until the eval rubric is signed at Gate 2. The discipline is structural, not aspirational.

Risk 3 — Gate fatigue

Eight gates feel like eight bottlenecks; teams collapse them to ship faster, defeating the purpose.

Mitigation. The PRS permits gate-flagging at PRS-3: only flagged outputs trigger review. The four pedagogical gates are non-negotiable; the four production gates are sampleable at higher PRS levels. The framework is *designed to scale review intensity with the PRS level in operation*.

Risk 4 — The autonomy-responsibility problem

Recommending PRS-4 in any context is a public claim about when humans can step back. If the recommendation is wrong, the consequences fall on learners.

Mitigation. PRS-4 is bounded by classification cap (§10) and by domain risk; AAFL never claims PRS-4 fits credentialing, classified, or high-consequence performance contexts. The framework is *opinionated about its own limits*. Publishing what AAFL won't do is the source of the framework's credibility, not its limitation.

Risk 5 — Translator-not-evangelist voice risk

A framework launch reads as evangelism by default. The temptation is to lead with productivity numbers and capability claims.

Mitigation. Lead with the gates and the caps, not with the productivity numbers. Publish what AAFL *won't* do, in writing, before publishing what it does.

Risk 6 — Citation-and-source integrity in the implementation pipeline

The framework itself is exposed to the agent-era failure mode it warns against: agents that produce plausible-but-wrong citations into AAFL-produced artifacts. A Source Truth gate (Gate 4) that does not specifically audit citation accuracy will let through fabricated DOIs, mis-attributed claims, and confabulated author names — exactly the failure mode Hicks et al. (2024) describe.

Mitigation. Treat citation-verification as a named Gate 4 responsibility in the practitioner checklist: every cited source must resolve to a real artifact at the URL/DOI given; author names, year, journal/publisher, and page numbers must match the source as published; primary-vs-secondary distinctions must be preserved. AAFL implementations should embed a citation-verification step in the Develop-phase agent workflow (against a canonical-source registry where one exists for the program's domain) and a sampling audit at the human Gate 4 review. This risk is enumerated explicitly because it is the framework's most self-reflexive vulnerability and the one most likely to damage the framework's own credibility if not operated against.

IP and authorship

AAFL is independent IP authored by Ruchir Bakshi. The framework is offered to the field for citation, critique, and operational use. Suggested citation: *Bakshi, R. (2026). AAFL: An Agent-Augmented Framework for Learning (v1.3). Zenodo.* https://doi.org/10.5281/zenodo.20319475 (concept DOI — always-latest; v1.3 frozen at https://doi.org/10.5281/zenodo.20319476). Engagement with the framework on cited claims is welcome and expected; the engagement principle is *cite, don't pitch*.

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§12. Limitations and Future Work

A framework that publishes its own caps owes the reader an account of its own limits. AAFL is a publishing-ready spec, not a validated artifact, and the gap between the two should be visible to anyone deciding whether to adopt it.

Limitations

- **Not yet operated end-to-end at PRS-3+.** The framework's PRS-1 and PRS-2 prescriptions are grounded in the author's federal practitioner observations across ACE-accredited programs (see §1.5 Method). The PRS-3 productivity band reported in §6 is *projected, not measured*; no published field test of AAFL at PRS-3 or PRS-4 exists at this writing.
- **Single-author synthesis without external review panel.** AAFL was developed by one practitioner-researcher (see §1.5). The framework has not been subjected to Delphi consensus with an independent expert panel, independent peer review of the integrated framework (individual sources are peer-reviewed; the integration is not), or formal user research with a representative practitioner sample. This whitepaper is offered to the field with the explicit invitation that external review and field testing become the next stage of development.
- **Federal proof-of-application reported as practitioner observation, not citable artifact.** The author's federal ID practice produced the operational observations on first-draft cycle compression reported in §1 and the PRS-1 / PRS-2 productivity bands reported in §6. The underlying internal deliverables are not citable in public-facing AAFL artifacts. A reader who wishes to audit those observations independently does not have a published artifact to consult.
- **PRS productivity bands are practitioner-level observations, not controlled-study results.** The numerical ranges reported in §6 are anchored to published generative-AI productivity studies (Brynjolfsson et al., 2023; Dell'Acqua et al., 2023) where the empirical literature supports the anchor, and labeled as practitioner observation where the author's federal experience is the source. No row in §6 should be read as the output of a controlled experiment on AAFL itself.
- **Scope explicitly excludes K-12 and adult-non-workplace learning.** AAFL is designed for workplace performance contexts (federal training, corporate L&D, higher-ed-to-workplace pipelines). The gate taxonomy and the construct-validity dimension are transferable to K-12, but the PRS-caps-by-classification framing does not map cleanly to FERPA / COPPA regulatory regimes. K-12-specific application is outside the framework's claim space.
- **The pedagogical-vs-production split is a *theoretical commitment*.** The §5 task assignments (which rows sit in the agent column and which in the human-only column) are versioned. The commitment is to the discipline of asking *which decisions belong to humans regardless of capability*, not to a permanent row assignment. Field experience may require updates; see §11 Risk 4 and §5 "When the line moves."

- **The framework's workplace-performance organizing outcome is *instrumental*.** This is a scope choice, not a defect, but readers from the non-instrumentalist tradition in education (Bayne et al., 2020) will rightly observe that not all educational contexts are appropriately measured against performance outcomes. AAFL's scope claim is that workplace-performance contexts are the right home for instrumental measurement (people are paying for training to get better at their work); the framework takes no position on the non-workplace contexts where Bayne et al.'s critique applies. The non-instrumentalist tradition's deeper point — that teaching is constitutively relational and that reducing it to optimization targets is a category error in some contexts — is acknowledged as a limit on AAFL's scope, not as a critique the framework dissolves.

Future work

- **Field tests at PRS-3 on bounded scopes.** Two natural targets: one credentialing program (where PRS-2 is the appropriate cap and gate-discipline can be measured) and one CUI training program on approved federal-tenant infrastructure (where PRS-3 becomes available and gate-flagging behavior can be characterized). Where venue choice permits, field-test reports should be deposited to EdArXiv to maximize subsequent citability and field engagement.

- **External expert review panel.** A bounded Delphi or modified-Delphi exercise with five to ten federal ID and AI-in-education practitioners would test the framework's specifics against multiple independent expert perspectives, and surface row-level disagreements with the pedagogical-vs-production split that the single-author synthesis cannot.

- **Comparative practitioner study against FRAME-using practitioners.** AAFL maps decision authority; FRAME™ maps capability risk. A comparative study of practitioners using both, one, or neither would test the operational complementarity claim made in §9.

- **Open-source the eval rubric template and gate-checklist as living artifacts.** The companion tactical assets (the HITL Decision-Gate Practitioner Checklist and the AAFL Eval Rubric Template) are presently bundled with the framework spec. Releasing them as separately versioned artifacts under the same © Ruchir Bakshi 2026 attribution would let the field test, fork, and feed back into them without waiting for a v1.4 framework release.

- **K-12-specific application paper.** The gate taxonomy and construct-validity dimension are transferable; the PRS-caps-by-classification framing is not. A separate paper extending AAFL to K-12, re-anchored in FERPA/COPPA rather than FedRAMP/NIST 800-171, would address the audience explicitly excluded here without compromising the workplace-performance commitment of the present framework.

The framework's structure is designed to tolerate row-level updates, comparative validation, and federation with adjacent published frameworks without architectural rewrites. The published spec is the starting point; field experience is the next contributor.

§13. Bibliography

Allen, M. W., & Sites, R. (2012). *Leaving ADDIE for SAM: An agile model for developing the best learning experiences*. ASTD Press.

Anthropic Engineering. (2026, January 9). *Demystifying evals for AI agents*. https://www.anthropic.com/engineering/demystifying-evals-for-ai-agents

Anthropic Research. (2024, December 19). *Building effective AI agents*. https://www.anthropic.com/research/building-effective-agents

Arya Mary, K. J., Bhaskar, D. K., Sinu, T. S., & Binu, V. P. (2026). *Agentic AI for education: A unified multi-agent framework for personalized learning and institutional intelligence* [Preprint] (arXiv:2604.16566). https://arxiv.org/abs/2604.16566 *(Source paper for the "AUSS — Agentic Unified Student Support System" architecture cited in §9.)*

ATD Research. (2025). *2025 State of the Industry Report*. Association for Talent Development. https://www.td.org/content/atd-blog/benchmarks-and-trends-from-the-2025-state-of-the-industry-report

Baldwin, T. T., & Ford, J. K. (1988). Transfer of training: A review and directions for future research. *Personnel Psychology*, *41*(1), 63–105. https://doi.org/10.1111/j.1744-6570.1988.tb00632.x

Bayne, S., Evans, P., Ewins, R., Knox, J., Lamb, J., Macleod, H., O'Shea, C., Ross, J., Sheail, P., & Sinclair, C. (2020). *The Manifesto for Teaching Online*. MIT Press.

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? 🦜 In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (pp. 610–623). Association for Computing Machinery. https://doi.org/10.1145/3442188.3445922

Bloom, B. S. (1984). The 2 sigma problem: The search for methods of group instruction as effective as one-to-one tutoring. *Educational Researcher*, *13*(6), 4–16. https://doi.org/10.3102/0013189X013006004

Branson, R. K., Rayner, G. T., Cox, J. L., Furman, J. P., King, F. J., & Hannum, W. H. (1975). *Interservice procedures for instructional systems development* (TRADOC Pam 350-30; Vols. 1–5). Florida State University, Center for Educational Technology. (ADDIE origin.)

Brinkerhoff, R. O. (2003). *The success case method: Find out quickly what's working and what's not*. Berrett-Koehler.

Brynjolfsson, E., Li, D., & Raymond, L. R. (2023). *Generative AI at work* (NBER Working Paper No. 31161). National Bureau of Economic Research. https://doi.org/10.3386/w31161 *(Published 2025 in* Quarterly Journal of Economics*, 140(2), 889–942.)*

Burke, L. A., & Hutchins, H. M. (2007). Training transfer: An integrative literature review. *Human Resource Development Review*, *6*(3), 263–296. https://doi.org/10.1177/1534484307303035

CAST. (2018). *Universal Design for Learning Guidelines version 2.2*. https://udlguidelines.cast.org/

Chai, D. S., Kim, H. S., Kim, K. N., Ha, Y., Shin, S. S. H., & Yoon, S. W. (2025). Generative artificial intelligence in instructional system design. *Human Resource Development Review*. https://doi.org/10.1177/15344843251320256

Dakan, R., & Feller, J. (2023–2025). *The AI fluency framework*. https://aifluencyframework.org/ *(Anchoring resource for the Anthropic Academy courses* AI Fluency for Educators *and* AI Fluency: Framework & Foundations*.)*

Dane, A. V., & Schneider, B. H. (1998). Program integrity in primary and early secondary prevention: Are implementation effects out of control? *Clinical Psychology Review*, *18*(1), 23–45. https://doi.org/10.1016/S0272-7358(97)00043-3

Dell'Acqua, F., McFowland III, E., Mollick, E. R., Lifshitz-Assaf, H., Kellogg, K., Rajendran, S., Krayer, L., Candelon, F., & Lakhani, K. R. (2023). *Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality* (Harvard Business School Working Paper 24-013). https://www.hbs.edu/faculty/Pages/item.aspx?num=64700 *(Published 2026 in* Organization Science *as Dell'Acqua et al., 2026; DOI: 10.1287/orsc.2025.21838.)*

Dick, W., Carey, L., & Carey, J. O. (2014). *The systematic design of instruction* (8th ed.). Pearson. (Original 1978.)

Driscoll, M. P. (2005). *Psychology of learning for instruction* (3rd ed.). Pearson.

Edelson, D. C. (2002). Design research: What we learn when we engage in design. *Journal of the Learning Sciences*, *11*(1), 105–121. https://doi.org/10.1207/S15327809JLS1101_4

EleutherAI. (2024). *lm-evaluation-harness: A framework for few-shot evaluation of language models*. GitHub. https://github.com/EleutherAI/lm-evaluation-harness

Gagné, R. M. (1985). *The conditions of learning and theory of instruction* (4th ed.). Holt, Rinehart and Winston. (Original 1965.)

Gilbert, T. F. (1978). *Human competence: Engineering worthy performance*. McGraw-Hill. (Tribute edition, Pfeiffer / International Society for Performance Improvement, 2007.)

Gottfredson, C., & Mosher, B. (2011). *Innovative performance support: Strategies and practices for learning in the workflow*. McGraw-Hill.

Hardman, P. (2025). *Defining and navigating the jagged frontier in instructional design* [Substack post, October 2025]. https://drphilippahardman.substack.com/p/defining-and-navigating-the-jagged *(Public articulation of the FRAME™ framework cited in §9 as the closest published precursor to AAFL's pedagogical-vs-production task split.)*

Hicks, M. T., Humphries, J., & Slater, J. (2024). ChatGPT is bullshit. *Ethics and Information Technology*, *26*, Article 38. https://doi.org/10.1007/s10676-024-09775-5

Holmes, W., Bialik, M., & Fadel, C. (2019). *Artificial intelligence in education: Promises and implications for teaching and learning*. Center for Curriculum Redesign.

Holton, E. F., III. (1996). The flawed four-level evaluation model. *Human Resource Development Quarterly*, *7*(1), 5–21. https://doi.org/10.1002/hrdq.3920070103

Kirkpatrick, D. L., & Kirkpatrick, J. D. (2016). *Kirkpatrick's four levels of training evaluation*. ATD Press. (Original 1959.)

Kraiger, K., Ford, J. K., & Salas, E. (1993). Application of cognitive, skill-based, and affective theories of learning outcomes to new methods of training evaluation. *Journal of Applied Psychology*, *78*(2), 311–328. https://doi.org/10.1037/0021-9010.78.2.311

Liang, P., Bommasani, R., Lee, T., Tsipras, D., Soylu, D., Yasunaga, M., Zhang, Y., et al. (2023). Holistic evaluation of language models (HELM). *Transactions on Machine Learning Research*. https://crfm.stanford.edu/helm/

Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, *12*, 157–173. https://doi.org/10.1162/tacl_a_00638

Luckin, R. (2018). *Machine learning and human intelligence: The future of education for the 21st century*. UCL IOE Press.

Mager, R. F. (1962). *Preparing instructional objectives*. Fearon. (Third edition, 1997, Center for Effective Performance.)

Mayer, R. E. (Ed.). (2014). *The Cambridge handbook of multimedia learning* (2nd ed.). Cambridge University Press. https://doi.org/10.1017/CBO9781139547369

McKenney, S., & Reeves, T. C. (2013). Systematic review of design-based research progress: Is a little knowledge a dangerous thing? *Educational Researcher*, *42*(2), 97–100. https://doi.org/10.3102/0013189X12463781

McKenney, S., & Reeves, T. C. (2018). *Conducting educational design research* (2nd ed.). Routledge.

Merrill, M. D. (2002). First principles of instruction. *Educational Technology Research and Development*, *50*(3), 43–59. https://doi.org/10.1007/BF02505024

Messick, S. (1989). Validity. In R. L. Linn (Ed.), *Educational measurement* (3rd ed., pp. 13–103). American Council on Education / Macmillan.

Meyer, A., Rose, D. H., & Gordon, D. (2014). *Universal Design for Learning: Theory and practice*. CAST Professional Publishing.

Mollick, E. (2024). *Co-intelligence: Living and working with AI*. Portfolio. (Centaur / cyborg distinction; autonomy gradient.)

Moore, C. (2017). *Map it: The hands-on guide to strategic training design*. Montesa Press. https://blog.cathy-moore.com/action-mapping/ *(Action Mapping was introduced on the author's practitioner blog c. 2008 and consolidated in this volume.)*

National Institute of Standards and Technology. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100-1; E. Tabassi, lead author). U.S. Department of Commerce. https://doi.org/10.6028/NIST.AI.100-1

O'Donnell, C. L. (2008). Defining, conceptualizing, and measuring fidelity of implementation and its relationship to outcomes in K–12 curriculum intervention research. *Review of Educational Research*, *78*(1), 33–84. https://doi.org/10.3102/0034654307313793

Quality Matters. (2023). *QM higher education rubric* (7th ed.). MarylandOnline. https://www.qualitymatters.org/

Reigeluth, C. M. (Ed.). (1999). *Instructional-design theories and models, Volume II: A new paradigm of instructional theory*. Lawrence Erlbaum Associates. (Volume III, with A. A. Carr-Chellman, 2009; Volume IV, with B. J. Beatty & R. D. Myers, 2017, Routledge.)

Rossett, A., & Schafer, L. (2007). *Job aids and performance support: Moving from knowledge in the classroom to knowledge everywhere*. Pfeiffer.

Rummler, G. A., & Brache, A. P. (1990). *Improving performance: How to manage the white space on the organization chart*. Jossey-Bass. (Third edition, 2012.)

SAE International. (2021). *Taxonomy and definitions for terms related to driving automation systems for on-road motor vehicles* (J3016_202104). SAE International. https://www.sae.org/standards/content/j3016_202104/

Saghafian, S., & Idan, L. (2024). Effective generative AI: The human-algorithm centaur. *Harvard Data Science Review*, Special Issue 5 (Generative AI). https://hdsr.mitpress.mit.edu/pub/3rvlzjtw

Selwyn, N. (2019). *Should robots replace teachers? AI and the future of education*. Polity Press.

Smith, P. L., & Ragan, T. J. (2005). *Instructional design* (3rd ed.). Wiley.

Sweller, J. (2011). Cognitive load theory. In J. P. Mestre & B. H. Ross (Eds.), *Psychology of Learning and Motivation* (Vol. 55, pp. 37–76). Academic Press. https://doi.org/10.1016/B978-0-12-387691-1.00002-8

VanLehn, K. (2011). The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. *Educational Psychologist*, *46*(4), 197–221. https://doi.org/10.1080/00461520.2011.611369

Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes* (M. Cole, V. John-Steiner, S. Scribner, & E. Souberman, Eds.). Harvard University Press.

Watters, A. (2021). *Teaching machines: The history of personalized learning*. MIT Press.

Web Content Accessibility Guidelines (WCAG) 2.1. (2018, updated). *W3C Recommendation*. World Wide Web Consortium. https://www.w3.org/TR/WCAG21/

Williamson, B. (2017). *Big data in education: The digital future of learning, policy and practice*. SAGE Publications.

Wilson, H. J., & Daugherty, P. R. (2018, updated 2024). *Human + machine: Reimagining work in the age of AI*. Harvard Business Review Press.

Wood, D., Bruner, J. S., & Ross, G. (1976). The role of tutoring in problem solving. *Journal of Child Psychology and Psychiatry*, *17*(2), 89–100. https://doi.org/10.1111/j.1469-7610.1976.tb00381.x

Zhang, X., Zhang, C., Sun, J., Xiao, J., Yang, Y., & Luo, Y. (2025). *EduPlanner: LLM-based multi-agent systems for customized and intelligent instructional design* [Preprint] (arXiv:2504.05370). https://arxiv.org/abs/2504.05370

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Appendix A. Acronym Glossary

A consolidated reference for the acronyms used in this paper. Every acronym below is also spelled out at first body-text appearance where space permits. **AAFL** is pronounced *ay-eff-ell* (or, informally, *apple* with a soft P).

Acronym	Expansion
AAFL	Agent-Augmented Framework for Learning
ACE	American Council on Education
ADDIE	Analyze · Design · Develop · Implement · Evaluate (the canonical five-phase instructional design model)
AR	Augmented Reality
AUSS	Agentic Unified Student Support System (per Arya Mary et al., 2026)
CAST	Center for Applied Special Technology (publisher of the UDL Guidelines)
cmi5	Computer-Managed Instruction profile of xAPI
CUI	Controlled Unclassified Information
DDVR	Draft · Discern · Validate · Release — the four steps of the Translator's Loop, run by the human practitioner against every agent-produced artifact within each ADDIE phase
DL	Design Loop (taxonomy tag: tasks that compress ID cycle time)
DoD	Department of Defense
EAR	Export Administration Regulations
EdArXiv	Open-access preprint repository for education research (referenced in §12 Future Work as a preferred deposit channel for forthcoming field-test reports)
FedRAMP	Federal Risk and Authorization Management Program
FERPA	Family Educational Rights and Privacy Act
FRAME™	Framework for Responsible AI in Mapping the (Jagged) Frontier (Hardman)
HITL	Human-in-the-Loop
HPT	Human Performance Technology
ID	Instructional Design / Instructional Designer
ISD	Instructional Systems Design
ISPI	International Society for Performance Improvement
ITAR	International Traffic in Arms Regulations

Acronym	Expansion
JKO	Joint Knowledge Online (DoD learning platform)
L1–L4	Kirkpatrick's four levels of training evaluation (Reaction · Learning · Behavior · Results)
L&D	Learning and Development
LMS	Learning Management System
NICE	National Initiative for Cybersecurity Education (NIST framework)
NIST	National Institute of Standards and Technology
OPM	Office of Personnel Management
PL	Performance Loop (taxonomy tag: tasks that compress time-to-competency or improve transfer-to-job)
PRS	Proportional Restraint Scale — AAFL's four-level autonomy scale (PRS-1 Sandbox · PRS-2 Co-designer · PRS-3 Production Partner · PRS-4 Performance System)
QM	Quality Matters
SAM	Successive Approximation Model (Allen Interactions)
SCORM	Sharable Content Object Reference Model
SME	Subject Matter Expert
UDL	Universal Design for Learning
WCAG	Web Content Accessibility Guidelines (W3C)
xAPI	Experience API (formerly Tin Can API)

Note. "Section 508" is a section of the U.S. Rehabilitation Act, not an acronym; it is preserved as-written throughout the paper. The taxonomy column-tag prefixes **(PRS-3+)** and **(PRS-4)** in §5 are PRS-gate prefixes, not acronyms; see §5 (Reading the PRS prefixes).

Appendix B. Comparison of Published AI-Era ID Frameworks

This appendix substantiates the federal-differentiation claim made in §8 and §10 by tabulating the architectural features of the published AI-era instructional design frameworks reviewed in this work, plus the most-cited established models AAFL inherits from. The scan is not exhaustive; it is the working set against which AAFL positions itself, and it should be expanded as additional frameworks are published or surfaced.

The columns are deliberately narrow architectural questions, not value judgments. *Performance-anchored?* asks whether the framework treats workplace performance (Kirkpatrick L3/L4 territory) as the organizing outcome rather than learning attainment. *HITL gates?* asks whether the framework specifies named decision points where human judgment is irreducible. *Progression model?* asks whether it prescribes levels with stopping points. *Classification caps?* asks whether progression is tied to data-classification context (the federal-applicability question). *Eval discipline* names the framework's stated approach to assessing the work agents produce. *Primary contribution* names what the framework adds to the field.

Framework	Author / Year	Organizing outcome	Performance-anchored?	HITL gates?	Progression model?	Classification caps?	Eval discipline	Primary contribution
AAFL	Bakshi (2026)	Workplace performance (Kirkpatrick L3/L4)	**Yes**	**Yes — 8 named gates with pedagogical/ production split**	**Yes — PRS-1 → PRS-4 with prescribed caps**	**Yes — by data classification (the framework's distinctive federal claim)**	Eval-as-spec; blocking rubric authored at Gate 2 before Design or Develop runs	Integration of all four: gates + PRS + caps + eval-as-spec
ADDIE	Branson et al. (1975)	Learning attainment (implicit)	No (learning-attainment, implicit)	No — phases, not decision gates	No	No	End-of-phase evaluation (Phase E)	The five-phase ID spine that the field still reads in
FRAME™	Hardman (2025)	Capability-risk management	No (capability-risk)	No — risk zones, not decision gates	No	No	Task-specific quality-zone checks	Maps capability risk across the jagged frontier
Anthropic 4Ds	Dakan & Feller (2023–2025)	Practitioner AI fluency	No (practitioner fluency)	No — competencies, not gates	No	No	Diligence-as-audit (one of the four competencies)	Educator-facing AI fluency competencies
Chai et al.	Chai et al. (2025)	Phase-by-phase AI integration	No (phase integration)	No	No	No	Not framework-prescribed	Peer-reviewed academic anchor for AI-in-ID
EduPlanner	Zhang et al. (2025)	Iterative ID optimization	No (iterative optimization)	No — three-agent harness, not human gates	No	No	Evaluator agent inside the three-agent loop	Multi-agent academic system for ID drafting
AUSS	Arya Mary et al. (2026)	Personalization at student / educator / institutional levels	No (multi-level personalization)	No — four-module agents, not human gates	No	No	Per-module evaluation inside the agent architecture	Four-module agent architecture across levels
			Yes		No	No		

Framework	Author / Year	Organizing outcome	Performance-anchored?	HITL gates?	Progression model?	Classification caps?	Eval discipline	Primary contribution
Action Mapping	Moore (2008/2017)	Business performance (backward design)		Implicit — backward-design checkpoints			Performance observation against business goal	Backward-from-business-goal design discipline
5 Moments of Need	Gottfredson & Mosher (2011)	Workflow learning + performance support	**Yes**	No — moments, not decision gates	No	No	Workflow observation across the five moments	Workflow learning continuum from formal to performance support
Success Case Method	Brinkerhoff (2003)	Did the intervention move performance?	**Yes**	No — case-finding method, not gates	No	No	Targeted case interviews + impact attribution	Method for tracing whether the intervention actually moved performance

Reading the scan

The integration is the contribution. No single row above is unique to AAFL except the classification-caps column. The eight gates, the four-level PRS, the eval-as-spec discipline, the workplace-performance organizing outcome — each individually has precedent or near-precedent in the cited works. What AAFL claims is the *integration*: a framework that simultaneously specifies (a) HITL decision gates with pedagogical/production classification, (b) a four-level scale with prescribed caps, (c) caps that are *tied to data classification* rather than to capability or context-in-general, (d) an eval discipline that *blocks* the Develop phase until the rubric is signed, and (e) a governance layer that enforces all of the above. The scan shows no other reviewed framework occupies all five positions.

The federal-caps column is where AAFL is structurally distinct. FRAME™, the 4Ds, and Chai et al. are all classification-agnostic frameworks. They could be operated in any context — commercial, federal, K-12, civilian or military — without changing their prescriptions. AAFL is the only reviewed framework whose architecture changes prescriptions based on the data-classification context of the work: classified content authoring caps at PRS-2; CUI training development can reach PRS-3 with approved infrastructure; public-trust unclassified high-volume work can reach PRS-4. This is the question federal contracting officers ask first, and AAFL's caps-by-classification framing is the framework's answer.

The performance-anchored column shows where AAFL sits in the broader L&D tradition. Action Mapping, 5 Moments of Need, and Success Case Method are all performance-anchored — they share AAFL's organizing-outcome commitment. The agent-era frameworks (FRAME™, 4Ds, Chai et al., EduPlanner, AUSS) are *not* performance-anchored; they are oriented to capability risk, practitioner fluency, phase integration, optimization, or personalization. AAFL is positioned at the intersection of the two lineages: it inherits the performance-anchoring of the HPT tradition and the agent-era operating discipline of the contemporary AI-in-ID frameworks, integrated under one architecture. No other reviewed framework occupies both positions.

Scope limits. This scan covers published AI-era ID frameworks at the time of drafting (mid-2026). It does not exhaust the corporate-LMS-vendor space (Articulate, Synthesia, Vyond, etc., are tools — not frameworks — and live in the §1 capability claim, not here). It does not cover the broader AI-in-education academic literature beyond the AI-in-ID corner (Selwyn, Williamson, Holmes-Bialik-Fadel, Luckin, Bayne, Watters are engaged in §9 as critic-camp voices, not as competing frameworks). Additional frameworks should be added as they are surfaced; the comparative-scan structure is versioned with the framework.
